

RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments

Ricardo B. Sousa 

THESIS RESEARCH PLAN



Programa Doutoral em Engenharia Electrotécnica e de Computadores

Supervisor: Prof. António Paulo Moreira , PhD

Second Supervisor: Héber M. Sobreira , PhD

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Resumo

A operação de robôs móveis autônomos a longo-termo coloca novos desafios ao problema de localização e mapeamento simultâneo. Num ambiente estático, o robô necessitaria apenas de uma passagem pelo espaço para construir a respetiva representação do mapa, dado que o mapa permaneceria consistente com o local ao longo do tempo. Contudo, aplicações industriais e orientadas para serviços, entre outras, têm elementos móveis no ambiente ou ocorrem mesmo reconfigurações de elementos do local. De facto, o sistema autónomo deveria decidir como e quando actualizar o mapa. Consequentemente, um sistema de localização e mapeamento simultâneo adequado a ambientes dinâmicos deve ser capaz de perceber as mudanças do espaço à volta do robô e manter uma representação do mapa actualizada com o meio envolvente dos robôs. Esta proposta de investigação de tese de doutoramento, intitulada *RicoSLAM: Localização e Mapeamento a Longo-Termo em Ambientes Dinâmicos*, visa realizar localização e mapeamento simultâneo a longo prazo em ambientes interiores e dinâmicos com um robô móvel autónomo. A proposta de investigação coloca a hipótese de como a estimativa da dinâmica do ambiente como um valor contínuo e a integração da sua estimativa tanto nos processos de localização como de mapeamento deveria melhorar os resultados desses processos em meios dinâmicos. A estimativa da dinâmica incluirá possivelmente o estudo de valores anteriores da dinâmica, se e quando os pontos do mapa existem ou não ao longo do tempo, e também a consideração das propriedades dinâmicas de conjunto dos pontos do mapa em relação à sua vizinhança. Além disso, esta proposta de investigação pretende também calibrar automaticamente o sistema sensorial do robô a fim de melhorar o desempenho da localização e mapeamento do ambiente, com base na convergência das estimativas de movimento relativo do robô a partir dos diferentes sensores. Em geral, esta proposta de investigação prevê contribuições significativas para o actual estado da arte sobre o tema de localização e mapeamento a longo-termo, visando uma solução estável para o problema de localização e mapeamento num ambiente dinâmico.

Palavras-chave: localização e mapeamento simultâneo, robô móvel autónomo, autonomia a longo prazo, ambientes dinâmicos.

Abstract

The long-term operation of Autonomous Mobile Robots (AMR) poses new challenges to the Simultaneous Localization and Mapping (SLAM) problem. In a static scene, the robot would require just one pass through the environment to build the latter's map representation, given that the map would remain consistent with the scene throughout time. However, industrial and service-oriented applications, among others, have inherently moving elements in the scene or even occurring reconfigurations of elements the environment. Indeed, the autonomous system should decide how and when to update the map, the so-called stability-plasticity dilemma. Consequently, a SLAM system adequate for dynamic scenes should be able to perceive the changes in the environment and maintain an up-to-date map representation of the robots' surroundings. This PhD thesis research proposal, entitled *RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments*, aims to perform lifelong SLAM within dynamics indoor environments with an AMR. The research proposal hypothesizes how estimating the environment dynamics as a continuous value and integrating their estimation in both localization and mapping processes should improve the SLAM results in changing environments. The dynamics estimation will possibly include the study of the dynamics history, if and when the map points exist or not throughout time, and also the consideration of clustering properties of the map points relative to their neighborhood. Additionally, this research proposal also intends to calibrate automatically the robot's sensory system in order to improve the SLAM performance, based on the convergence of relative motion estimations of the robot's frame from the different sensor sources. Overall, this research proposal foresees significant contributions to the current state of the art on the topic of long-term SLAM, aiming for a stable solution for the problem of performing SLAM in a dynamic environment.

Keywords: Simultaneous Localization and Mapping (SLAM), Autonomous Mobile Robot (AMR), long-term autonomy, lifelong SLAM, dynamic environments.

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Abbreviations

- AI** Artificial Intelligence. 3
- AMR** Autonomous Mobile Robot. 1–4, 10, 22, 23, 25–27
- ARMA** Auto-regressive Moving Average Model. 3, 21–23
- ATE** Absolute Trajectory Error. 18, 19, 21
- BDVA** Big Data Value Association. 3
- BoW** Bag-of-Words. 20
- CAD** Computer-Aided Design. 18
- CAGR** Compound Annual Growth Rate. 1
- CLAIRE** Confederation of Laboratories for Artificial Intelligence Research in Europe. 3
- CML** Concurrent Mapping and Localization. 10
- CNN** Convolutional Neural Network. 2, 20, 23
- CRIS** Centre for Robotics in Industry and Intelligent Systems. 34
- DBN** Dynamic Bayesian Network. 13
- DEEC** Department of Electrical and Computers Engineering. 34, 35
- DEL** Dynamic Edge Link. 19
- DFR-SLAM** Dynamic visual Feature Removal Simultaneous Localization and Mapping. 18, 19
- DoF** Degrees of Freedom. 25, 28
- DSRM** Design Science Research Methodology. 4, 5
- EKF** Extended Kalman Filter. 13, 14
- ELLIS** European Lab for Learning & Intelligent Systems. 3
- EU** European Union. 39
- EurAI** European Association for Artificial Intelligence. 3
- euRobotics** European Robotics. 3
- FEUP** Faculty of Engineering of the University of Porto. 33–35, 37, 41
- FOV** Field-Of-View. 26, 33, 34
- FreME**n Frequency Map Enhancement. 3, 21–23
- FSH** Feature Stability Histogram. 17
- HMM** Hidden Markov Model. 21
- HOG** Histogram of Oriented Gradients. 21
- ICP** Iterative Closest Point. 12, 18, 19, 22
- IMU** Inertial Measurement Unit. 25
- INESC TEC** Institute for Systems and Computer Engineering, Technology and Science. 33, 34, 36, 39, 43
- iSAM** incremental Smoothing And Mapping). 15
- LC-CRF** Long-term Consistent Conditional Random Field. 18
- LiDAR** Light Detection And Ranging. 2, 12, 19, 20, 23, 26, 33, 39

- LTD** Long-Term Dynamics. 21
- LTM** Long-Term Memory. 3, 17, 20, 22
- MCL** Monte Carlo Localization. 18, 21, 22
- MLE** Maximum Likelihood Estimation. 15
- NDT** Normal Distribution Transform. 12, 19, 23
- NICP** Normal Iterative Closest Point. 12
- POV** Point-Of-View. 4
- PRISMA** Preferred Reporting Items for Systematic reviews and Meta-Analysis. 7–9, 16
- RBPF** Rao-Blackwellized Particle Filter. 13
- RMSE** Root Mean Square Error. 21
- RNN** Recurrent Neural Network. 20, 21
- ROS** Robot Operating System. 15
- SDF** Signed Distance Field. 18
- SDG** Sustainable Development Goal. 4
- SGC-VSLAM** Semantic and Geometric Constraints Visual Simultaneous Localization and Mapping. 21
- SIVO** Semantically Informed Visual Odometry. 20
- SLAM** Simultaneous Localization and Mapping. 1–5, 7–16, 22–24, 26–30, 33, 35, 37–43
- SM** Sensory Memory. 17
- SRIDA** Strategic Research, Innovation and Deployment Agenda. 3, 4
- STD** Short-Term Dynamics. 21
- STM** Short-Term Memory. 3, 17, 22
- TOF** Time-Of-Flight. 26
- TORO** Tree-based netwORk Optimizer. 15

Chapter 1

Introduction

This research aims to contribute to the progress of the current state of the art on **Simultaneous Localization and Mapping (SLAM)**. This work focuses on challenges inherent to the long-term operation of **Autonomous Mobile Robots (AMRs)** while performing SLAM in dynamic indoor environments. The main goals are to maintain a stable and up-to-date representation of the environment and, consequently, improve the localization performance of **AMRs**. In this context, the proposed approach should develop new algorithms and take advantage of emerging technologies to overcome the challenges found through the research work.

This chapter is divided into five sections. Section 1.1 contextualizes this research work and describes the primary challenges related to long-term **Simultaneous Localization and Mapping (SLAM)**. Section 1.2 presents the motivations that led to the formulation of this research plan. Then, Section 1.3 establishes the goals for the PhD thesis. Section 1.4 details the research methodology that will support the research work. Finally, Section 1.5 defines the structure of this document.

1.1 Context

Nowadays, logistic and retail companies, warehouses, automotive and aerospace industries, agriculture businesses, and healthcare institutions are using **AMRs** to automate tasks such as transportation and security monitoring. The tasks automation improves operational efficiency, flexibility, and productivity achieving higher efficiencies and lower production costs [1]. Consequently, these improvements contribute to the growth of the global economy and increase the global interest in the **AMR** market. Indeed, a study elaborated by AgileIntel Research [2] estimates the value of the global **AMR** market at US\$2,525.66 million in 2021 and is estimated to grow at a **Compound Annual Growth Rate (CAGR)** of 15.4% between 2022 and 2029, as illustrated in Figure 1.1. The conclusions of the study highlight the importance of **AMRs** for global economics while also forecast an increasing demand for **AMR** systems in the following years from 2022 [2].

An **AMR** requires a representation of its surroundings by extracting meaningful data from its sensors to localize itself relative to the environment [3]. **SLAM** addresses this problem by incorporating the robot state estimation concurrently with the mapping process to build a representation of the environment perceived by the robot. The state is described by the robot's pose (position and orientation), even though other variables may be included in the state, such as velocity and calibration parameters. As for the environment representation, the mapping process originates a map incrementally built when exploring unknown areas or refined on passages through known locations in order to describe the environment in which the robot operates [4].

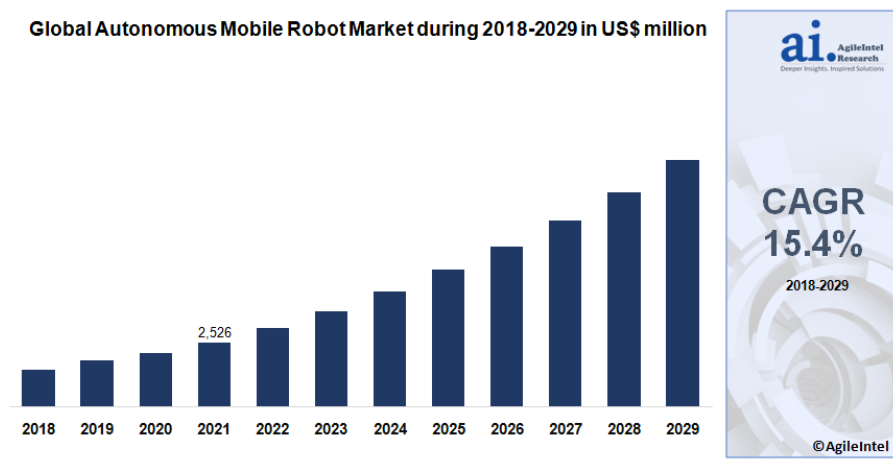


Figure 1.1: Global **Autonomous Mobile Robot (AMR)** market estimated growth by AgileIntel Research during 2018–2029 in US\$ million [2].

In a static scene, the robot would only need to map once because the map would remain consistent with the environment throughout time. However, industrial locations, agricultural sites, warehouses, and service-oriented applications such as shopping centers and healthcare institutions have inherently moving elements in the scene (humans, objects), reconfiguration of the environment (reorganization of machines, patient beds), and appearance variations (lighting, weather, seasonal, or daytime changes). The deployment of **AMRs** in these varying conditions are a challenge for the **SLAM** system, where the system should decide how (consider the most current state, the evolution of the environment dynamics, only the most static changes?) and when to update the map (when the variations occur, after a certain time?). This challenge is also known as the stability-plasticity dilemma, where a lifelong **SLAM** system should both adapt to new environment changes and preserve old states over time [5]. Furthermore, the map would grow indefinitely when gathering new information from the environment without any map management system. This ever-growing problem poses another challenge for the **SLAM** system in the long-term due to the limited computational resources of the mapping vehicle. Indeed, the map size should be dependent on the environment area and not on the trajectory length of the mapping process nor operation time, only growing when the robot would explore unknown locations [6].

Currently, the literature is predominantly focused on the appearance invariance challenge of long-term localization and mapping, especially in outdoor environments, given the recent growing interest in autonomous vehicles. Although vision sensors are greatly affected by appearance variations, **Convolutional Neural Network (CNN)**-based features extracted from the visual images are being used for localization in challenging outdoor scenes, given their more discriminative properties compared to handcrafted features [7–12]. The current research also uses other sensor modalities, such as **Light Detection And Ranging (LiDAR)** [13–16] (also known as 3D laser scanners) and Radar [16, 17], to overcome some limitations of vision sensors. While visual data is affected by illumination changes of day-night situations or changing weather and seasonal conditions, laser-based localization does not suffer from illumination variance. Even so, fog or direct light can still affect the laser readings. As for radar, the sensor is invariant to both lighting and weather changes. Still, noisy measurements affect the performance of radar-based **SLAM** in long-term scenarios [16]. Moreover, the existent literature also evaluates the long-term stability of features to only using stable ones for localization and mapping, using as scoring functions predictors such as number of observations and concentration ratio [18, 19].

Several works already focus on the ever-growing problem of continuously performing **SLAM**. The concept of mutual information from information theory is the most common methodology to determine which nodes to remove from the map for the graph formulation of **SLAM** [6, 20–23]. However, this methodology is mostly intended for offline execution (*e.g.*, between running sessions) due to its additional computational cost when operating online. Other methodologies to manage the size of the graph evaluate outdated information between current view and the map [24, 25], the spatial density of nodes [26, 27], and the node time recency [28, 29]. In feature-based **SLAM**, management policies such as counting matching observations of features [30, 31] and using multiple predictor functions [18, 19, 32, 33], similarly to appearance invariance methods, also try to constraint the number of mapped features to only be dependent on the explored environment area.

As for dealing with dynamic environments, an intuitive approach is to distinguish between static and dynamic observations, favoring the stable ones for localization, and having two maps representing the most stable and the more dynamic representations of the environment [24, 34, 35]. Inspired by the human memory multi-store model of Atkinson and Shiffrin [36], **Short-Term Memory (STM)** and **Long-Term Memory (LTM)** maps try to capture at least two different levels of dynamics [5, 30, 31]. Furthermore, the rejection of dynamic objects or features for localization and mapping to only consider stable observations is performed in the literature by methods such as the YOLO network [35, 37] and pixel-wise semantic segmentation [38–40]. The methodologies mentioned previously for dealing with dynamic environments model the environment dynamics as static (permanent change), semi-static (short-term static object or static at the current observation instant), or dynamic (moving object in the scene). In contrast, assuming the independence of observations, Markov-based processes can be used to model and predict a value for the environment dynamics [41, 42]. However, these models are usually intended for offline usage between sessions and the assumption of independent observations may not be valid because static and dynamic objects may influence the dynamics of their surroundings. Although the **Frequency Map Enhancement (FreME)** [43] algorithm estimates online the dynamics through spectral analysis, the method only assumes periodic changes in the environment. The **Auto-regressive Moving Average Model (ARMA)** [44] models both aperiodic and periodic changes but, still, its offline operation does not allow an online estimation of the environment dynamics.

1.2 Motivation

A joint partnership between major European activities for **Artificial Intelligence (AI)** – **Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE)**, **European Lab for Learning & Intelligent Systems (ELLIS)**, and **European Association for Artificial Intelligence (EurAI)** –, **Data – Big Data Value Association (BDVA)** –, and **Robotics – European Robotics (euRobotics)** – released a **Strategic Research, Innovation and Deployment Agenda (SRIDA)** [45] in September 2020. This agenda identifies key and critical technical areas for the future of Robotics, among them are dealing with unpredictable and unknown environments and irreversible actions. Indeed, in many service applications, surrounding environments of autonomous systems usually involve the interaction with humans. Furthermore, bad policies can bring autonomous systems to unrecoverable states from which learning is no longer possible, requiring the environment to be reset to start the process again. For example, the influence of dynamic elements in the process of **SLAM** with **AMRs** can make the environment representation saved in the robot outdated and unusable due to changes in the scene without any module to update the map accordingly to the environment changes. An outdated map representation would

required a specialized operator to remap the environment, possibly redefining the trajectories if needed, and then restart the autonomous system operation.

Subsequently, the interpretation and processing of dynamic changes in the environment by **AMRs** would enable the maintenance of an up-to-date map representation of the robots' surroundings. Robot systems would be able to better understand their operating environment, which may include humans, while also improving the efficiency and safety of these systems. For these outcomes, **SRIDA** [45] also identifies technology and perception enablers such as robot systems being able to interpret sensor data in real-time – particularly in complex environments or fusion of multi-modal sensor data (vision, ranging, acoustic) to increase self-awareness of those systems –, matching local **Point-Of-View (POV)** with world views, and full 3D perception systems and sensors able to decompose and interpret whole scenes in real-time to 4D.

1.3 Goals

The main scope of this research work, *RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments*, is performing lifelong **SLAM** within dynamic environments with an **AMR**. The focus is developing map maintenance algorithms capable of handling dynamic changes and moving elements in the operating area of autonomous robot systems. Thus, the main goals of this research proposal aiming to contribute to the progress of the current state of the art on **SLAM** are the following ones:

- modeling of environment dynamics perceived by different sensors (2D and 3D lasers, depth data from RGBD cameras) in the map representation;
- account the estimated values of the environment dynamics incorporated in the map for both localization and mapping processes;
- achieve long-term localization and mapping in dynamic environments.

Within the scope of the 2030 Agenda for sustainable development agreed by 195 nations in September 2015 [46], this research work falls under the following **Sustainable Development Goal (SDG)**:

- SDG8: Decent work and economic growth;
- SDG9: Industry, innovation and infrastructure;
- SDG12: Responsible Consumption and Production.

1.4 Research methodology

The definition of a base research methodology is crucial to formulate a robust research strategy for obtaining successful results. In engineering disciplines, the **Design Science Research Methodology (DSRM)** is considered a valid and valuable method to obtain applicable and effective problem solutions incrementally [47]. Thus, this PhD thesis will follow a **DSRM** strategy as the one presented in Peffers *et. al.* [47] and illustrated in Figure 1.2. **DSRM** incorporates the principles, practices, and procedures to carry out research projects. The methodology consists on the following nominal sequential processes:

1. Problem identification and motivation;
2. Definition of the objectives for a solution;
3. Design and development;
4. Demonstration;
5. Evaluation;

6. Communication.

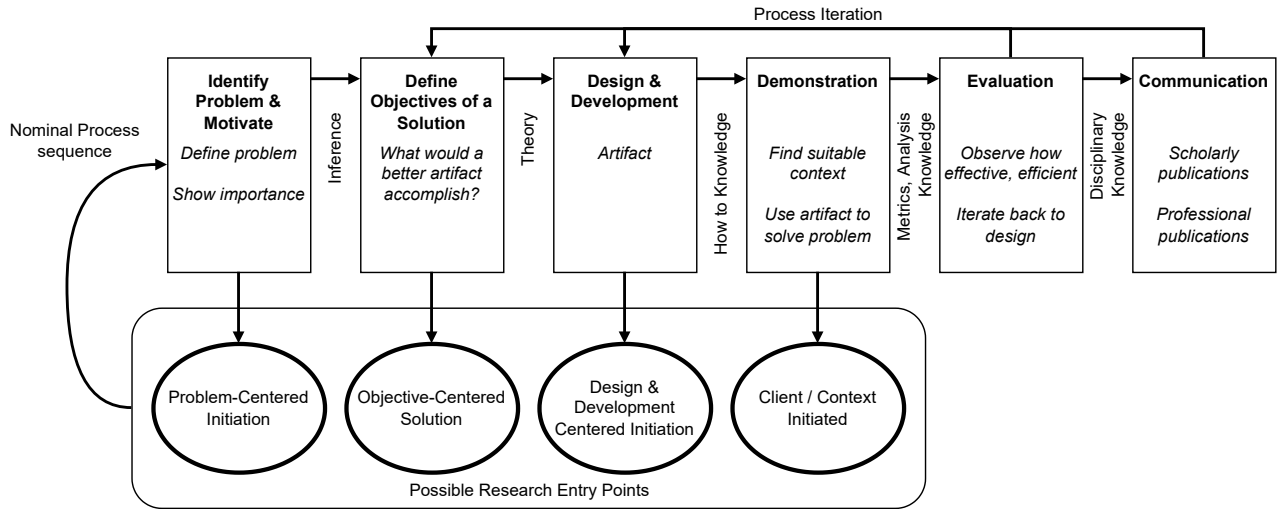


Figure 1.2: Process model of the **Design Science Research Methodology (DSRM)** [47].

The first process defines the specific research problem and justifies the value of a solution and its relevance to the scientific community. The second step defines both quantitative (based on key performance indicators and in comparison to existing solutions) and qualitative (functional description of the solution) goals from the problem definition. The following step focuses on design and development activities for creating an artifact, defining its desired functionality and architecture and then actually creating it. In the scope of this PhD thesis, an artifact can be models, algorithms, methodologies, among others. The *Demonstration* activity uses the artifact to solve one or more scenarios or case studies of the problem. Then, the next process is the observation and measurement of how well the artifact supports the solution of the problem. The *Evaluation* can be a comparison of the artifact's functionality with the objectives from the second activity in the **DSRM** model, simulations, or feedback from the end-user of the solution. Lastly, the *Communication* step focuses on the scientific dissemination of the problem and its importance, the artifact, its novelty and utility, the methodology for its design, and its effectiveness to researchers and other relevant audiences such as specialized operators. Although Peffers *et al.* [47] structures the **DSRM** model in a nominally sequential order, the methodology does not obligate the researchers to always proceed sequentially from the first to the sixth activity. Indeed, Peffers *et al.* [47] defines possible research entry points for the researchers to begin and then move outward.

1.5 Document organization

The remaining of this document is organized as follows. Chapter 2 presents a literature review on concepts and technologies related to the topics of **SLAM** and long-term localization and mapping in dynamic environments. Chapter 3 provides the problem statement and the main goals and scope of this research proposal. The statement describes the problem and formulates the research questions that should be answered during the execution of this research work and possible hypotheses that support potential answers to those questions. Chapter 4 describes preliminary work and developments. Finally, Chapter 5 outlines the work plan defining the tasks and milestones predicted for the research and the respective timeline. This chapter also discusses the scientific dissemination methodology that will be followed during the execution of the research plan.

Chapter 2

Literature Review

The preprint of a systematic literature review on long-term localization and mapping following the **Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA)** [48] guidelines is available in Appendix A of this document. The study precedes the elaboration of this thesis research plan and includes an analysis of 142 literature works for discussing and analyzing methodologies focused on appearance invariance, dynamic elements, map sparsification and multi-session techniques, and other computational concerns. Additionally, the study discusses commonly used public datasets and the characteristics of the experimental data used by the authors of the included works in the review while presenting the most used evaluation metrics. However, given the main scope of this research work on lifelong **SLAM** in dynamic environments, this chapter focus mainly on the discussion of localization and mapping algorithms in scenes with dynamic elements presented in the systematic review. This chapter also formulates the **SLAM** problem and its most common solutions.

Hence, this chapter is divided into four sections. Section 2.1 summarizes the methodology of the systematic review (further detail available in Appendix A) on long-term localization and mapping and overviews the included literature works in terms of publication year with an emphasis on **SLAM** in dynamic environments. Section 2.2 formulates the traditional **SLAM** problem and presents its three most common solutions: map matching, particle filter, and pose graph implementations. Then, Section 2.3 discusses literature works related to long-term **SLAM** in dynamic environments based on the findings presented in Appendix A.

2.1 Literature overview

2.1.1 Review methodology

The literature review is an important process in any research project for providing a comprehensive summary of previous research on a specific topic. This process aims to identify the main contributions, research gaps, define current challenges and future directions in the scope of a research topic. However, the review methodology should be supported on relevant and rigorous frameworks for synthesizing the findings of the literature works.

A systematic literature review uses explicit, rigorous, and reproducible systematic methods to synthesize the findings of works related to a particular research question, topic area, or phenomenon of interest. This type of review assures the quality and trustworthiness of the review’s findings by presenting a complete, organized, and summarized analysis of all works considered while allowing other researchers to replicate or update the review. The most common standard for performing systematic reviews is the **PRISMA** [48] statement. Even though the

PRISMA [48] guidelines have been designed originally for evaluating the effects of health interventions, the checklist items of the methodology are general and applicable to other subject areas. Therefore, the literature covered in this chapter and extensively discussed in Appendix A follow a systematic methodology based on the **PRISMA** [48] guidelines. This methodology is composed by the following steps:

1. definition of eligibility criteria;
2. conceptualization of the search strategy;
3. selection process:
 - (a) identification;
 - (b) screening;
 - (c) quality assessment.
4. data extraction;
5. synthesis and analysis of the included works.

The first step defines the eligibility criteria to determine the eligible studies for the selection process. In the systematic review on long-term localization and mapping, the criteria considered are mostly related with the type of paper – indexation in scientific venues, subject area, reject short papers and gray, secondary, and tertiary literature, exclude dataset papers – and availability – full-text accessibility and language. Additionally, a coverage (exclude odometry-only approaches) and scope (not consider different or non related subjects) criteria are exclusion criteria focused on the scientific contributions of the works. Next, the search strategy conceptualization defines as data sources the databases ACM Digital Library [49], Dimensions [50], IEEE Xplore [51], INSPEC [52], Scopus [53], and Web of Science [54]. The date of the last full inquiry in the systematic review was May 17, 2022. However, for the discussion of long-term **SLAM** in dynamic environment in Section 2.3, recent works between the previous data until October 25, 2022, will be included for the analysis. As for the base string used to inquiring the data sources considering the title, abstract, and keywords search fields is the following one: `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong)`. The selection process starts by identifying the records in the different sources using the previous base string, possibly rejecting duplicate records (Parsifal [55] was the tool used for identifying duplicates) and also based on the eligibility criteria defined previously. The screening phase also excludes works based on the eligibility criteria by reading the title and abstract of the publications. Then, the quality assessment establishes another set of criteria focused on the scientific contribution of the eligible records. The assessment of eligible records rejects works based on works with no compliance in certain criteria (*e.g.*, non detailed methodology or results sections, and works not having long-term experiments or not considering localization nor mapping problems). Finally, the data extraction process allows the retrieval of information for the synthesis and analysis of the included works while also rejecting articles with extended versions to improve the discussion and singularity of the methodologies proposed in the literature. Figure 6 illustrates the **PRISMA** [48] flow diagram for summarizing the selection process. Further details on the systematic methodology followed for synthesizing and analysis literature works on long-term localization and mapping can be found in the Section 3 *Methodology* of the Appendix A.

2.1.2 Results overview

The relevance of the long-term localization and mapping topic can be evaluated by the evolution of the number of publications. Figure 2.2 presents this evolution using the Title, Abstract, and Keywords (authors and indexed

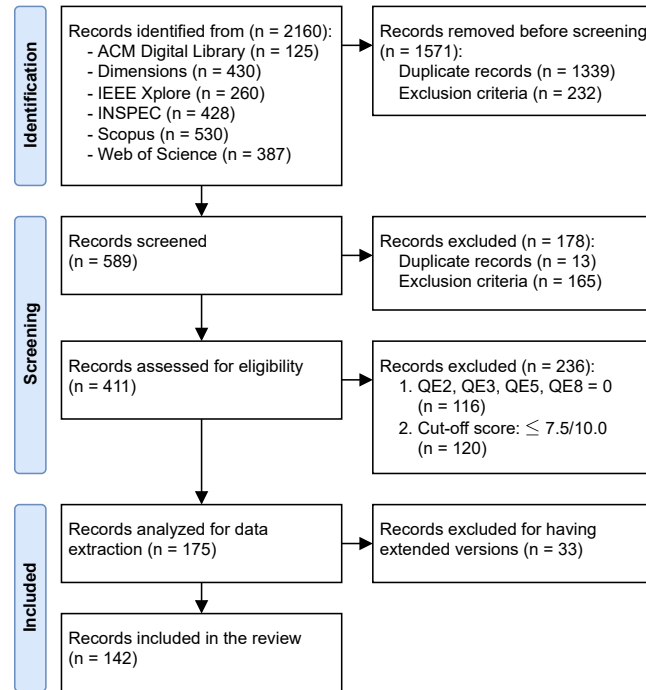


Figure 2.1: Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) [48] flow diagram for the selection process of the systematic review on long-term localization and mapping. Date of the last full inquiry: May 17, 2022.

ones) as search fields when inquiring in three different bibliographic databases: INSPEC [52], Scopus [53], and Web of Science [54]. Even though the inquiry only represents the identification phase of the selection process in a systematic review (*i.e.*, applying the search strategy to all data sources), still, the search results allow having a perception of published works related to the long-term **SLAM** and the interest of the scientific community in the topic. In Figure 2.2a, the search string used for the inquiry is the same one used in the systematic review: `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong)`. This string does not focus on any particular challenge related to long-term localization and mapping. In contrast, Figure 2.2b presents the evolution of the number of publications reducing the scope of the search results to the challenge of localization and mapping in dynamic environments, the same one directly related to the goals of this research proposal. This scope reduction is accomplished by using the following search string: `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*)`. The part of the query added to the original one (`dynamic* OR mov*`) intends to limit the search results to works related to dealing with dynamic or moving elements in the scene.

When analyzing Figure 2.2, firstly, the graph in Figure 2.2a shows an increasing interest of the scientific community in the topic of long-term localization and mapping. Since 2009, the number of publications are predominantly more than 10 per year. From that year onwards, the graph has an almost linear tendency reaching a maximum of 74, 76, and 62 published records on the years of 2019, 2021, and 2021 respective to the databases INSPEC [52], Scopus [53], and Web of Science [54], respectively. The tendency shows that long-term **SLAM** is gaining relevance throughout the years, supporting the importance and relevance of the topic for the scientific community. This result is in conformity to the analysis of the year of publication presented in the systematic review for the 142 literature works included in the paper (see Section 4 *Results Overview* of the Appendix A).

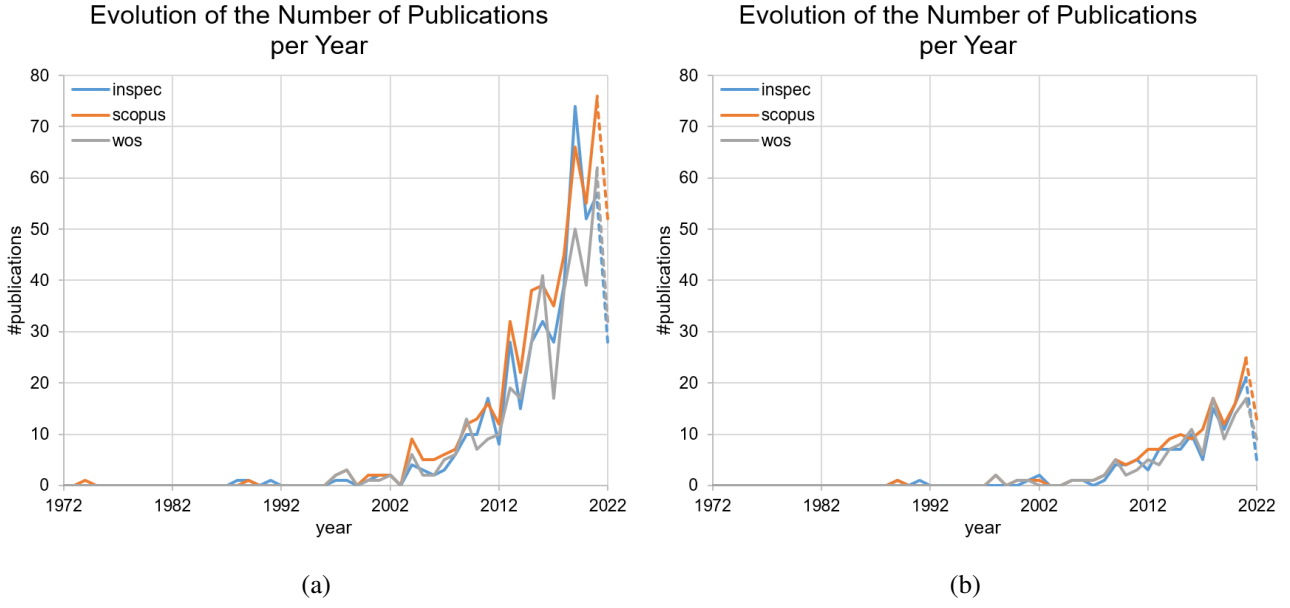


Figure 2.2: Evolution of published records per year in the INSPEC [52], Scopus [53], and Web of Science [54] databases on long-term localization and mapping: (a) (robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong); (b) (robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*). Date of the last full inquiry: October 25, 2022.

Similarly, the graph shown in Figure 2.2b indicates a growing interest in the topic of this research proposal, *i.e.*, long-term localization and mapping in dynamic environments. The topic also seems to have gained relevance from 2009 onwards, achieving a maximum of 21, 25, and 17 published works in 2021 respective to the databases INSPEC [52], Scopus [53], and Web of Science [54], respectively. Although the number of publications related to dynamic environments (Figure 2.2b) is lower than the ones associated with long-term SLAM (Figure 2.2a), the growing interest in solving the challenges of performing localization and mapping in dynamic environments demonstrates the relevance of this research proposal.

2.2 Simultaneous Localization and Mapping (SLAM)

Next, this section focuses on defining the SLAM problem and discussing the three main techniques for SLAM: map matching, particle filter, and pose graph. Given the focus of this thesis research plan on maintaining an up-to-date representation of the environment when operating AMRs in dynamic surroundings, the algorithms to be developed should not only localize the robot in the current map but also update the latter accordingly to the dynamics of the scene elements. Thus, the problem of long-term localization and mapping in dynamic environments is intrinsically related to SLAM.

2.2.1 Problem introduction

SLAM is one of the most fundamental problems in robotics. This problem, also previously known as **Concurrent Mapping and Localization (CML)** in the literature, may be formulated by the Definition 2.2.1. When the robot does not have access to a representation of the environment nor know its pose, SLAM enables an AMR to

build a map of its surroundings while simultaneously localizing itself relative to this map. **SLAM** may also be utilized for building the map used in applications assuming a static scene, even if the map is only continuously updated upon the mapping process [56].

Definition 2.2.1 ([3, p. 349]). “...the robot must not only create a map, but it must also do this while moving and localizing to explore the environment. In the robotics community, this is often called the *Simultaneous Localization and Mapping (SLAM)* problem.”

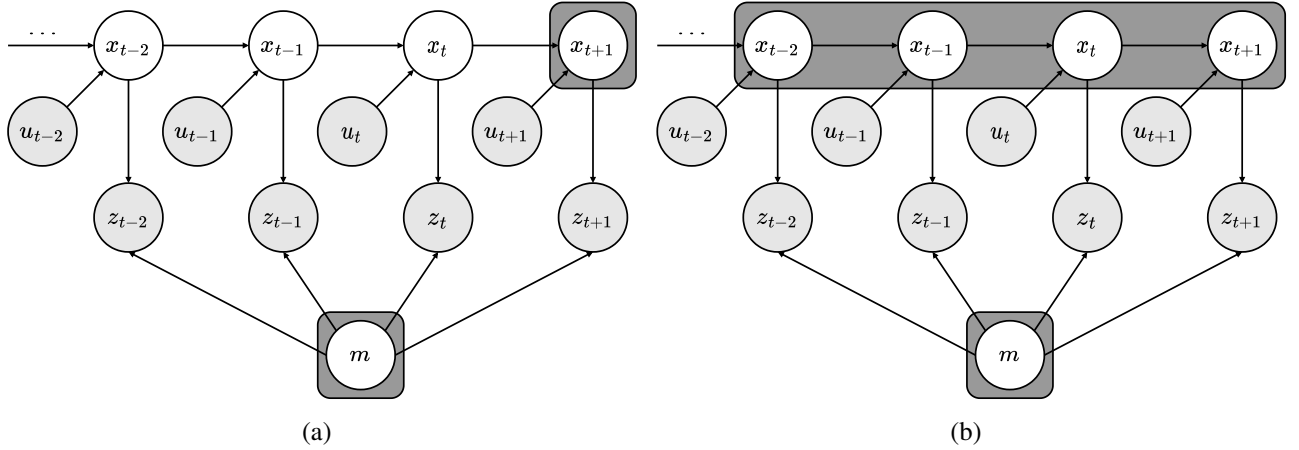


Figure 2.3: Graphical model of the **Simultaneous Localization and Mapping (SLAM)** problem: (a) online SLAM; (b) full SLAM [56].

Two main formulations define **SLAM** from a probabilistic perspective, also illustrated graphically in Figure 2.3. The first represented in Figure 2.3a is online **SLAM**, where the algorithm estimates the posterior over the momentary pose (x_t at a time instant t) along with the map (m), as defined in Equation 2.1, provided the sensor measurements ($z_{1:t}$) and the control variables ($u_{1:t}$). The online term is due to only involving the estimation of variables that persist at the current time instant t . Several algorithms incrementally implement online **SLAM**, *i.e.*, discarding past measurements and controls once they have been processed. The other probabilistic formulation is called full **SLAM** and illustrated in Figure 2.3b, given the goal to compute a posterior over the entire path ($x_{1:t}$) along with the map, as formulated in Equation 2.2, instead of just the current pose (x_t). The difference between online and full **SLAM** is that the former results in an integration of the past poses from the full **SLAM** problem [56].

$$p(x_t, m \mid z_{1:t}, u_{1:t}) \quad (2.1)$$

$$p(x_{1:t}, m \mid z_{1:t}, u_{1:t}) \quad (2.2)$$

In the case of the algorithm also estimating correspondences between measurements and map landmarks, the explicit inclusion of the correspondence variables (c_t or $c_{1:t}$) in the estimation problem may be useful for reasoning if the perceived feature is the same one as previously detected and already represented in the map or if it is a new landmark. Consequently, Equations 2.3 and 2.4 formulate the posterior estimation for online and full **SLAM**, respectively [56].

$$p(x_t, m, c_t \mid z_{1:t}, u_{1:t}) \quad (2.3)$$

$$p(x_{1:t}, m, c_{1:t} \mid z_{1:t}, u_{1:t}) \quad (2.4)$$

Although computing a full posterior captures all the information to be known about the map and the pose or the path, this estimation is practically unfeasible due to the high dimensionality of the continuous parameter space and the large number of discrete correspondence variables [56]. So, next, the discussion on scan matching, particle filter, and pose graph implementations of **SLAM** focuses on the practical aspects of performing continuous localization and mapping.

2.2.2 Scan matching

A basic approach for solving the **SLAM** problem is continuously matching the sensor scans to the map learned until the current instant. An example is the HectorSLAM [57] algorithm, where the block diagram describing that system is shown in Figure 2.4. This algorithm is considered to be online **SLAM** due to only estimating the current pose and the map. Also, the method integrates incrementally the sensor readings to allow discarding those in subsequent iterations. Furthermore, HectorSLAM [57] aligns the beam endpoints of a 2D **LiDAR** with the current map using a matcher inspired by an alignment algorithm based on computer vision [58]. The matcher estimates the homogeneous transformation between the sensor scan and the map for updating the map with the latest scan and also update the current estimation of the robot pose. Although the matcher is specific to the HectorSLAM [57] algorithm, it does not invalidate using other matchers, such as the PerfectMatch [59], Iterative Closest Point (ICP) [60], Normal Iterative Closest Point (NICP) [61], Normal Distribution Transform (NDT) [62], or others, for obtaining the transformation between the 2D **LiDAR** scan and the map.

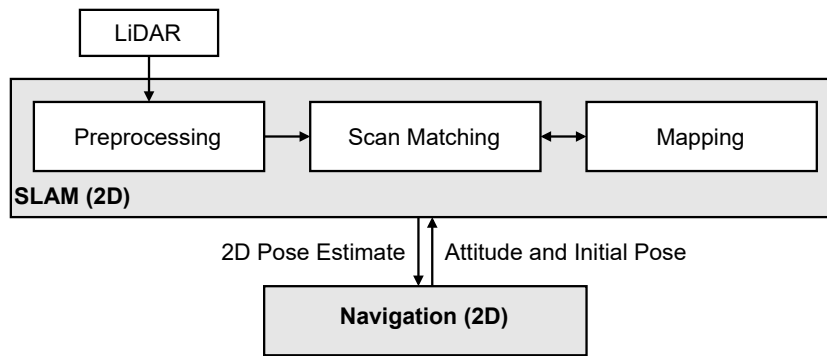


Figure 2.4: Block diagram of the mapping and navigation systems of Hector **SLAM**. Dashed lines represent optional information [57].

The main advantage of scan matching **SLAM** algorithms is the lower use of computational resources compared to more complex approaches. These algorithms do not require an exhaustive pose search or data association search with previous scans (the matching with all preceding scans is implicit in matching the current one with the map) [57]. However, as in the case of HectorSLAM [57], a pure scan matching approach does not implement loop closure (see Definition 2.2.2). This limitation does not allow correcting the map when recognizing a previously mapped place, not having the consequent reduction in the uncertainty of the map estimate.

Definition 2.2.2 ([3, p. 188]). “The problem of location recognition is also called loop-detection, because a loop is a closed trajectory of a vehicle that returns to a previously-visited point. The problem of minimizing the error at the loop closure is instead called loop-closing.”

2.2.3 Particle filter

The graphical model of the SLAM problem shown in Figure 2.3 is equivalent to a discrete Dynamic Bayesian Network (DBN) (see Definition 2.2.3). A sampling-based inference algorithm for DBNs is the particle filter. However, these filters suffer from the problem of dimensionality, scaling exponentially [63]. The assumption of conditional independence between map measurements allows using Rao-Blackwellized Particle Filters (RBPFS) for SLAM. These filters use particles to represent the posterior probability over some variables (e.g., the robot path, correspondence variables) and Gaussians (or other parametric probabilistic distribution functions) to represent all other variables (e.g., the feature positions) [56]. This marginalization of some variables implemented by RBPFS improves the computational efficiency compared to standard particle filters, reducing the space size for the sampling step [63].

Definition 2.2.3 ([64]). “A convenient way to describe this structure is via the Dynamic Bayesian Network (DBN)... A Bayesian network is a graphical model that describes a stochastic process as a directed graph. The graph has one node for each random variable in the process, and a directed edge (or arrow) between two nodes models a conditional dependence between them”

An example of a RBPFS-based SLAM is FastSLAM [65, 66]. This algorithm decomposes the SLAM problem into a robot localization one and the estimation of map landmarks conditioned on the robot pose estimate. The decomposition and factoring of the posterior probability ($p(x_t, m \mid z_{1:t}, u_{1:t}, c_{1:t})$) followed by the FastSLAM [65, 66] algorithm is shown in Equation 2.5. The algorithm assumes the conditional independence of the features (m_n) given the robot path ($x_{1:t}^{[N]}$), i.e., knowing the location of one feature does not add any additional information about the location of other features if the robot path is known. Moreover, Figure 2.5 represents the composition of particles implemented in FastSLAM. Each particle k contains an estimate of the robot path until the current time instant t and a set of Extended Kalman Filters (EKFs) with a mean ($\mu_j^{[k]}$) and covariance ($\Sigma_j^{[k]}$) of the estimation, one for each feature $j = 1..N$ represented in the map m . The basic steps of the FastSLAM [65, 66] algorithm are the following ones [56]:

- for each particle $k = 1..M$:
 1. **retrieval**: retrieve a pose $x_{t-1}^{[k]}$ from the particle set;
 2. **prediction**: sample a new pose $x_t^{[k]} \sim p(x_t \mid x_{t-1}^{[k]}, u_t)$ using the probabilistic motion model of the robot (the improved version, FastSLAM 2.0 [66], incorporates also the sensor measurements z_t^i in this step);
 3. **measurement update**: for each observed feature obtained from a measurement z_t^i , identify the correspondence j in the particle map, and incorporate z_t^i into the corresponding EKF, updating the mean ($\mu_j^{[k]}$) and covariance ($\Sigma_j^{[k]}$) of the corresponding map feature accordingly;
 4. **importance weight**: compute the importance weight $w^{[k]}$ for the new particle.
- **resampling**: sample, with replacement, the M particles, where the probability of sampling being proportional to $w^{[k]}$, to incorporate the measurements z_t^i into the robot pose $x_t^{[k]}$ (due to only considering the most recent control u_t in the previous steps, only in FastSLAM 1.0 [65]).

	robot path	feature 1	feature 2	feature N
Particle $k = 1$	$x_{1:t}^{[1]} = \{(x \ y \ \theta)^T\}_{1:t}^{[1]}$	$\mu_1^{[1]}, \sum_1^{[1]}$	$\mu_2^{[1]}, \sum_2^{[1]}$	$\dots \mu_N^{[1]}, \sum_N^{[1]}$
Particle $k = 2$	$x_{1:t}^{[2]} = \{(x \ y \ \theta)^T\}_{1:t}^{[2]}$	$\mu_1^{[2]}, \sum_1^{[2]}$	$\mu_2^{[2]}, \sum_2^{[2]}$	$\dots \mu_N^{[2]}, \sum_N^{[2]}$
		$\vdots$		
Particle $k = M$	$x_{1:t}^{[M]} = \{(x \ y \ \theta)^T\}_{1:t}^{[M]}$	$\mu_1^{[M]}, \sum_1^{[M]}$	$\mu_2^{[M]}, \sum_2^{[M]}$	$\dots \mu_N^{[M]}, \sum_N^{[M]}$

Figure 2.5: Composition of particles in FastSLAM [65, 66] in terms of path estimate and the set of estimators and associated covariances for individual feature locations [56].

$$\begin{aligned}
 p(x_t, m \mid z_{1:t}, u_{1:t}, c_{1:t}) &= p(x_t \mid z_{1:t}, u_{1:t}, c_{1:t}) p(m \mid x_t, z_{1:t}, c_{1:t}) = \\
 &= p(x_t \mid z_{1:t}, u_{1:t}, c_{1:t}) \prod_{n=1}^N p(m_n \mid x_t, z_{1:t}, c_{1:t})
 \end{aligned} \tag{2.5}$$

The adaptation of FastSLAM [65, 66] to occupancy grid-based **SLAM** is implemented by the GMapping [67] algorithm. This adaptation requires the modification of three modules relative to FastSLAM [65, 66]. First, GMapping [67] considers the occupancy grid as the map representation for the **SLAM** problem, where each particle contains a grid map. Next, the computation of a measurement likelihood ($p(z_t \mid x_t^{[k]}, m_{t-1}^{[k]})$) uses the beam endpoint sensor model, characteristic of range-bearing sensors such as 2D lasers, instead of integrating the measures into an **EKF** as in FastSLAM [65, 66]. This model considers the individual scan beams independent from each other, using the distance between the beam endpoint and the closest obstacle in the map to that point to compute the beam's individual likelihood. Lastly, the update of each particle's map ($p(m_t^{[k]} \mid z_t, x_t^{[k]}, m_{t-1}^{[k]})$) computes a new occupancy grid map. This update estimates the probability of each individual cell, assuming the independence of each other, using a binary Bayes filter and the inverse model of the sensor measurements [56].

A key advantage of particle filter-based **SLAM** is the decisions on data associations being made on a per-particle basis, instead of the most likely one as in scan matching or **EKF** approaches. This characteristic represents inherently multi-hypotheses associations in the filter, making it more robust to data association problems. Another advantage is particle filters being capable to model non-linear robot motions without requiring model approximations. However, a disadvantage of particle filters is relative to loop closure detection. Even though these filters are able to detect previously visited places, unlike scan matching approaches such as HectorSLAM [57], this place recognition ability is only maintained through the diversity in the particle sets, possibly maintaining larger sets and requiring more computational resources [56].

2.2.4 Pose graph

Another solution for the **SLAM** problem leveraging its graphical model (illustrated in Figure 2.3) is pose graph **SLAM**, also known as GraphSLAM. This optimization-based solution consists of two subsystems, as in filter-based approaches such as particle filters and **EKF**. In filter-based **SLAM**, the filters implement a prediction

stage usually based on the vehicle's motion model, and then, the correction phase updates the predicted state by matching the sensor observation to the map. As for optimization-based solutions such as the pose graph formulation, a first module also known as the front-end is responsible to identify constraints of the problem based on sensor data by finding correspondences between new observations and the current map. Then, a second subsystem known as the back-end computes or refines the poses of the graph and the map minimizing the coherence errors as a whole. This second module usually requires optimization algorithms such as Gauss-Newton or Levenberg-Marquardt for minimizing the objective function [68]. However, there are available in the literature optimization more specific for the pose graph formulation, *e.g.*, **Tree-based netwORk Optimizer (TORO)** [69], **incremental Smoothing And Mapping (iSAM)** [70, 71], or g^2o [72].

Furthermore, the edge connection in GraphSLAM between vertexes x_i and x_j is illustrated in Figure 2.6. Considering z_{ij} as a virtual measurement obtained from raw sensor data between those two vertexes, let $\bar{z}_{ij}$ and Ω_{ij} be the mean and the information matrix of the virtual data, respectively. This virtual measurement actually represents the transformation that maximizes the overlap between the observations acquired from the pose nodes i and j . The log-likelihood l_{ij} (see Definition 2.2.4) of the measurement z_{ij} can be defined as in Equation 2.6, where $\hat{z}_{ij}(x_i, x_j)$ is the prediction of the virtual measurement given the configuration of the nodes x_i and x_j . Then, the configuration of nodes $x^* = (x_1, \dots, x_T)^T$ can be found formulating a **Maximum Likelihood Estimation (MLE)** problem that minimizes the negative log-likelihood function $F(x)$ (see Equation 2.7) of all the observations, seeking to solve the Equation 2.8 [64]. For exemplification purposes, two possible examples of virtual measurements are wheeled odometry and 2D laser scan matching data. Both cases would define the prediction of the measurement as $\hat{z}_{ij} = x_j - x_i$. The virtual measurement would be $z_{ij} = x_j - g(x_i, u_{ij})$ for wheeled odometry, where $g(x_i, u_{ij})$ represents the relative transformation between x_i and x_j estimated by the odometry estimator, while using the wheeled odometry data u_{ij} between those two nodes. As for the scan matching case, the measurement would be $z_{ij} = h(z_i, z_j)$ representing the matching between the laser scans z_i and z_j acquired from the nodes x_i and x_j , respectively.

Definition 2.2.4 ([73, p. 224]). “...since the natural logarithm is a one-to-one function on $\mathbb{R}^+$, no information about θ is lost when the log-likelihood function $l(\theta) = \ln[p(\theta)]$ is used in place of $p(\theta)$. Moreover, for many probability models, it is easier to work with the log-likelihood function than with the likelihood function...”

$$l_{ij} \propto [z_{ij} - \hat{z}_{ij}(x_i, x_j)]^T \Omega_{ij} [z_{ij} - \hat{z}_{ij}(x_i, x_j)] \quad (2.6)$$

$$F(x) = \sum_{i,j} e_{ij}^T \Omega_{ij} e_{ij} \quad (2.7)$$

$$x^* = \underset{x}{\operatorname{argmin}} F(x) \quad (2.8)$$

A pose graph **SLAM** implementation is the **SLAM Toolbox** framework [74], also available in the official navigation stack of the **Robot Operating System (ROS)** [75, 76] (version ROS2). This toolbox is intended for 2D **SLAM** using a 2D laser scanner in conjunction with wheeled odometry obtained from the wheel encoders of the robot's motors. The optimization framework used for the pose graph back-end is the Ceres Solver [77]. Although **SLAM Toolbox** [74] states having lifelong mapping capabilities, these are only relative to being able to update existing maps and serialize the data for use in other mapping sessions, not for dynamic environments or appearance invariance.

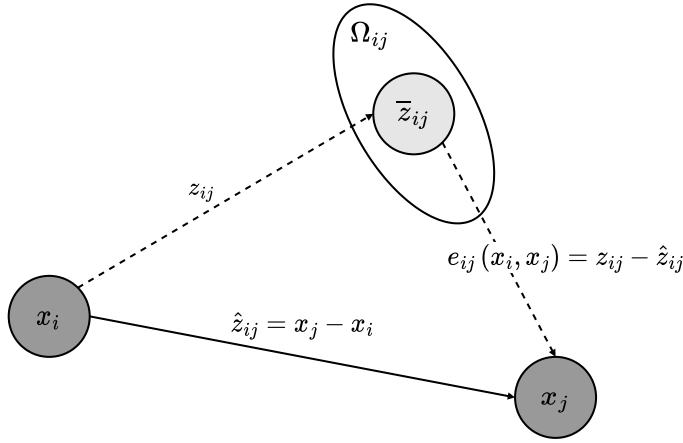


Figure 2.6: Edge connection in the graph formulation of the **SLAM** problem between vertexes x_i and x_j originated from a measurement z_{ij} [64].

Overall, a key difference between pose graph **SLAM** compared to other solutions for the **SLAM** problem, such as filter and scan matching approaches, is that the former has access to the full data when building the map, being able to apply improved linearization and data association techniques. Indeed, in GraphSLAM, all data can be used to linearize and calculate correspondences, *i.e.*, can revise past data association and linearize more than once [56]. This key difference is pointed as a possible reason to graph **SLAM** tending to give better results than filters, where the latter is more subject to linearization issues [68]. However, the pose graph formulation shows a linear growth over time in the graph's size requiring techniques for constraining the graph size (see Section 1.1), and thus, limit the computational resources required for continuously performing GraphSLAM [56].

2.3 Long-term SLAM in dynamic environments

In terms of works in the literature for long-term **SLAM**, as previously mentioned, Appendix A includes a systematic literature review on that topic following the **PRISMA** [48] guidelines. The literature review goes over several challenges related to long-term autonomy having the date of the last full inquiry for identifying records in the data sources considered in the study on May 17, 2022. Given the focus of this research proposal on the challenge for performing **SLAM** in dynamic environments, this section focus on the included works of the review study categorized as dynamic in the data extraction phase of the methodology. Furthermore, another full inquiry was performed on October 25, 2022, to include more recent works published or made available online between the date considered in the review up to the latter one. As a result, after executing all stages of the review methodology described in Appendix A and focusing only on dynamic-related works, another five records were found in the literature ([78–82]). However, given that the work [81] is an extension of [83], as assumed in the review, only the extended version will be considered for the discussion.

So, next, this section analyzes literature works focused on modeling and identifying dynamic elements in the environment within the scope of long-term **SLAM**, *i.e.*, how the methods identify these elements and handle them for long-term localization and mapping. The discussion on dynamics modeling is organized into the following topics: specific map representations used to model or deal with dynamic elements in the scene,

identification of dynamic elements matching the current observation to the current map, future prediction of dynamic properties of scene elements, and semantic identification of dynamic objects.

2.3.1 Map representation

Inspired by the human memory, Dayoub *et. al.* [30] and Bacca *et. al.* [31] adapt the multi-store model of Atkinson and Shiffrin [36] for robot mapping. This model divides the memory into three stores: **Sensory Memory (SM)** to save the perceived information, **STM**, and **LTM**. Three mechanisms move information between memories: selective attention for **SM** to **STM**, rehearsal to commit information from **STM** to **LTM** or which one is forgotten, and the retrieval mechanism to move unused information from **LTM** back to **STM**. Dayoub *et. al.* [30] implements two types of state machines for rehearsal and retrieval mechanisms of the **STM** and **LTM**. In rehearsal, a **STM** feature moves closer to **LTM** or moves back to the initial state (or forgotten if already in that state) when observed consecutively or if not, respectively. Similar for retrieval, where a feature in **LTM** moves to the initial state or closer to forget if observed in the current view or not. Consequently, **LTM** and **STM** save the most static and dynamic features based on their observability in the current view, respectively. In a changing environment, the method decreased the localization failure rate compared to a static view. Instead of using a state machine, Bacca *et. al.* [31] implements a **Feature Stability Histogram (FSH)** depending on the feature observability to distinguish between **STM** and **LTM** features using k -means clustering. This modification allows that an input feature in **SM** can bypass **STM** and become part of **LTM** depending on the feature strength. The method was able to filter out pedestrians from 2D laser and camera data and achieved a more accurate representation of the environment compared to a static approach.

Although Biber and Duckett [5] does not adopt specific memory mechanisms, they implement **STM** and **LTM** maps. The method implements a dynamic map as a set of local maps, each maintaining submaps representing different timescales. The timescale parameter of each submap determines probabilistically when to add samples from 2D laser scans. The dynamics of the environment are represented by using 5 different timescales, where the smaller one ($\sim 3.1s$) represents an **STM** map updated at every instant and the other 4 timescales ($\sim 0.43runs$, $0.43days$, $3.1days$, and $13.5day$) are **LTM** submaps updated after each session or daily. Instead of only localizing on the **LTM** maps as Dayoub *et. al.* [30] and Bacca *et. al.* [31], Biber and Duckett [5] selects the best representation of both **STM** and **LTM** maps that best explain the sensor data. In a 5 weeks experiment, the localization with a dynamic map improved while a static representation degraded over time, while the timescales led to static parts as walls emerging in **LTM** and dynamic elements disappearing from **STM** maps.

Similar to the works [30] and [31], two maps can represent a more stable and a more dynamic representations of the environment. In Walcott-Bryant *et. al.* [24], an active map represents the most current state of the environment, including parts that did not change from previous passes and objects added to the environment. A dynamic map only saves the points of a 2D laser scan that changed over time. Wang *et. al.* [34] uses a tracking map with short-term static points and a long-term map only containing long-term static points identified by a semantic segmentation module with ORB-SLAM2 [84]. Also, Zhu *et. al.* [35] creates offline a semi-dynamic map and a static one, where the former has semi-dynamic objects (parked cars in a parking lot environment) and the latter has both static and semi-dynamic objects. The main goal of representing two different dynamics is usually to favor the most stable one in the long-term. Walcott-Bryant *et. al.* [24] and Wang *et. al.* [34] only use static parts in the most current representation of the environment (active and the long-term maps, respectively) for localization. In the experiments, the approach [24] showed that their method was able to identify static

parts, although it was affected by false positives and negatives, and by the blur effect in the 2D grid map. The method [34] improved the **Absolute Trajectory Error (ATE)** in a dynamic environment over ORB-SLAM2 [84] and DynaSLAM [85]. As for Zhu *et. al.* [35], its **Monte Carlo Localization (MCL)** framework (equivalent to particle filter-based localization) reduces the weight corresponding to observations of moved semi-dynamic objects. The method improved the localization in a parking lot compared to a standard **MCL**.

As for a multi-hypothesis representation, Nielsen and Hendeby [79] propose a framework for tracking the landmarks positions using Markov chains to model the transition between positions. The method [79] groups the positions hypotheses based on the landmark's signature, where the position with the highest probability within a group according to incoming observations is selected as the active one. The hypothesis likelihood depends on the matched measurements to the landmark in the map and also on the number of missed detections, *i.e.*, landmarks with a probability of being seen in the current pose but not represented in the observations. Although the method [79] maintained an up-to-date representation of the actual landmark positions in simulated experiments, the experiments assumed always uniquely identifiable landmarks.

2.3.2 Map matching

Environment dynamics can be identified by comparing the current observation to the map. Assuming a prior vector map as a permanent map, the work of Biswas and Veloso [86] determines the probability of observed features being long-term ones by the 2D laser scan-to-map matching distance. Short-term features are determined by the scan-to-scan matching distance, while the remaining ones are considered dynamic features and not considered for localization. Compared to **MCL** with a static map and to Tipaldi *et. al.* [41], Biswas and Veloso [86] obtained lower localization error in a parking-lot environment. However, the method [86] would not handle semi-static changes. Instead of using a permanent map, Zhang *et. al.* [87] maintains a **Signed Distance Field (SDF)** representation based on a prior occupancy map. The proposed approach [87] rejects dynamic points identified by range flow and updates the **SDF**-based map with semi-static changes observed in the scan-to-map difference. Compared to **MCL** in a semi-static environment, Zhang *et. al.* [87] had lower pose errors and an improved representation of the environment. Similarly, Fu *et. al.* [82] uses **SDFs** in the mapping process, proposing the PlaneSDF representation where a scene is a set of planes and each one has associated a **SDF** volume. This volume describes the geometric details of the objects attached to its associated plane. Then, a change detection algorithm performs a 2D height map comparison and a 3D geometric validation between two PlaneSDF-based maps to compute a change mask. However, the method [82] failed to detect changes in ambiguous situations such as object occlusions in the height comparison direction. Boniardi *et. al.* [25] detects semi-static changes leveraging the **ICP** scan-to-map consistency and a **Computer-Aided Design (CAD)** prior of the environment, and updates the map accordingly. The method [25] was capable of maintaining a consistent map when dealing with substantial reconfiguration of the environment. Du *et. al.* [88] minimizes the Gibbs energy defined on the proposed **Long-term Consistent Conditional Random Field (LC-CRF)** for detecting dynamic points, considering that these points have often a large reprojection error in frame-to-map matching and points tend to have the same dynamic properties as the neighbor ones. In a dynamic scene, **LC-CRF** [88] achieved lower **ATE** than ORB-SLAM [89]. Unlike **LC-CRF** [88], the **Dynamic visual Feature Removal Simultaneous Localization and Mapping (DFR-SLAM)** proposed by Liu *et. al.* [78] does not look for similar dynamic properties in the features neighbor. Instead, the method [78] assumes that static features have consistent motion over long time and dynamic ones have different motion patterns, labeling them using the proposed motion

consensus filtering algorithm. **DFR-SLAM** [78] formulates binary classification problem based on a graph-cut optimization framework, assigning penalties to connected visual features between different instants that do not share the same label. The method [78] had lower **ATE** than ORB-SLAM2 [84] in dynamic scenarios.

Furthermore, Pan *et. al.* [90] and Ding *et. al.* [91] leverage clustering properties of the observations evaluating the observations count. The approach [90] segments the points of a 3D **LiDAR** point cloud into different clusters assuming that dynamic points do not appear frequently in the same place. The map only considers clusters that appear in same location more than 10 times. As for Ding *et. al.* [91], the method builds on the assumption that dynamic and static parts of the environment have a clustering property relative to its neighbors (similar to Du *et. al.* [88]). The number of observations in different sessions combined with its consistency relative to its neighbors determine if a map point is static throughout the sessions. Both representations of the environment in Pan *et. al.* [90] and Ding *et. al.* [91] were stable to structural changes in the environment.

The concept of ray tracing is also used by the included works to handle dynamic changes. Lázaro *et. al.* [92] uses ray tracing to exploit the free space information. When comparing two 2D point clouds from a viewpoint, the ray tracing evaluation identifies new objects added to the scene (observed point closer to viewpoint than the old one) and outdated information (observed point further way), allowing the identification of dynamic changes and having an up-to-date representation for localization. Given that ray tracing in 3D is expensive in terms of memory and requires dense map representations, Pomerleau *et. al.* [93] uses directly the sparse point cloud from a 3D laser. The map points are associated with each single reading in small conical apertures in spherical coordinates, updating the observed points closer than the mapped ones and the further ones are left untouched. The approximation of ray tracing results are used to update the probability of points in the map to be dynamic, based on a Bayesian approach. The probability of being dynamic can be used in **ICP** to not trust dynamic points and indeed, in the experiments, the work [93] had a more precise and cleaner map of the environment than using a standard **ICP** matching. Instead of weighting the map points, An *et. al.* [94] proposes the **Dynamic Edge Link (DEL)** to model the dynamics in the edges of a pose graph instead on the data itself. The observation of moving obstacles between two poses change the weight of the respective edge, decrease gradually the weight until not detecting the obstacle. Integrating **DEL** in an exploration scheme, nodes with a edge weight average lower than a certain threshold, meaning frequent moving obstacles or changed structure near that node, are not considered for exploration due to the robot may be unable to move to that position.

Although the standard **NDT** representation does not model free space, the works of Einhorn and Gross [95, 96] and Saarinen *et. al.* [97] use **NDT** with occupancy maps to model explicitly the free space and adopt exponential weighted moving average and covariance for new measurements having an higher influence than old ones. Einhorn and Gross [96] proposes a generic 2D/3D mapping using **NDT** and occupancy maps. The hit cells considering the current observation are updated incrementally with exponential weighting. The other cells along the sensor beam potentially empty are updated using the standard update rule of occupancy maps based on the log-odds of the occupancy value and on the inverse range sensor model. Instead of using the standard occupancy map update, the sensor model in Saarinen *et. al.* [97] depends on the inconsistency between observation and map. Also, the occupancy value describes the confidence of the **NDT** based on past observations. As for Einhorn and Gross [96], the method defines two probabilities for the occupancy map: occupancy and statically occupied, where the first is updated based on the sensor model (2D/3D generic beam sensor), and the second one is adapted slowly to high probability for static objects in the environment. The statically occupancy probability follows the proposed ad-hoc model that is parameterized to control how fast the static occupancy probabilities are adapted, depending also on the occupancy probability itself. The works [95] and

[96] were able to handle semi-static and dynamic changes having a consistent and up-to-date representation of the environment, while the approach [97] favored long-term static structures in dynamic environments.

2.3.3 Dynamic objects detection

In terms of detecting dynamic objects, Yue *et. al.* [37] proposes a collaborative dynamic mapping for detecting humans using visual and thermal images and a 3D **LiDAR**. The YOLOv3 algorithm extracts the bounding boxes from the images relative to humans. The 3D point cloud projection onto the images allows the creation of a static point cloud for localization and mapping of each robot by filtering out the points corresponding to humans. In the experiments, the dynamic objects removal generated a more accurate relative transformation of the collaborative maps compared to not removing those objects. Zhu *et. al.* [35] also uses YOLOv3 to extract bounding boxes of dynamic classes (parked cars) from a RGB image, and the projection of 3D **LiDAR** points allows the creation of the static and semi-dynamic maps required for localization. Even though the method [35] does not discard the dynamic objects, the localization module reduce the importance of weight of the corresponding observations. Instead of using visual data, Sun *et. al.* [98] adapts the PointNet network for object recognition (pedestrian, cyclist, car, or background) to classify the scan points of a 3D **LiDAR**. The proposed Recurrent-OctoMap maintains the occupancy and semantic information, where the latter information specifies the cell semantic state and the probability of the prediction. The transition between states is learned by a **LTM Recurrent Neural Network (RNN)**. In a long-term experiment, the method [98] was able to improve its 3D semantic map compared to a standard Bayes update.

Moreover, pixel-wise semantic segmentation is another way to identify dynamic objects. Additionally to the semantic-descriptor proposed by Singh *et. al.* [38] describing the spatial distribution of semantic entities in the scene, the method sets lower weights to features detected on sky and dynamic classes (person, car, *etc.*) from the semantic segmentation of images using EdgeNet. The proposed approach [38] obtained higher accuracy when compared to SeqSLAM [99], FAB-MAP [100], and **Bag-of-Words (BoW)** [101]-based place recognition methods in a highly dynamic outdoor experiment. Instead of identifying object classes, Song *et. al.* [39] proposes the MD-Net **CNN** to segment a grayscale image into unstable, static, and moving pixel points, only using static points for localization. The localization error was reduced compared to not estimated the pixel dynamic attribute. Ganti and Waslander [40] proposes the **Semantically Informed Visual Odometry (SIVO)** to improve the performance of ORB-SLAM2 [84] by using the Bayesian neural network SegNet for segmentation and computation of the network uncertainty. SegNet is trained to distinguish different object classes for identifying dynamic objects (sky, car, truck/bus, person/rider, motorcycle/bicycle, and void) from static ones (road, traffic sign, building, wall/fence, pole, vegetation, sidewalk, traffic light, and terrain). Only static keypoints that reduce the most of the state's uncertainty (considering the network uncertainty) are considered as input for ORB-SLAM2 [84]. **SIVO** [40] was able to remove uninformative and dynamic keypoints from the current frame. However, the rejection of potential dynamic objects without verifying if they are moving reduced the localization performance of the module in certain scenarios.

Dynamic objects identification can be improved by verifying geometric constraints. Bescos *et. al.* [85] proposes DynaSLAM, using multi-view geometry verification for improving semantic labeling of potential dynamic classes obtained from a Mask **RNN**. Although DynaSLAM [85] outperformed ORB-SLAM2 [84] in highly dynamic scenarios, its performance reduces in slower dynamics. Similarly, Wang *et. al.* [34] also uses a geometric verification for improving labeling of dynamic objects, the latter obtained by a ResNet-based

network. The verification is based on the reprojection error. The method [34] improved the ATE over DynaSLAM [85] and ORB-SLAM2 [84] in a scenario with movable objects. Guo and Liu [80] also uses a Mask RNN network for pixel-wise segmentation and object instance labels, similar to DynaSLAM [85]. Rather than using multi-view geometry verification just for improving semantic labeling, this verification is used in [80] also for detecting untrained moving objects. The approached [80] outperformed DynaSLAM [85] in terms of ATE in dynamic scenes. Instead of using semantic segmentation, the **Semantic and Geometric Constraints Visual Simultaneous Localization and Mapping (SGC-VSLAM)** [102] proposed by Yang *et. al.* uses YOLOv3 to extract bounding boxes of dynamic objects for also improving ORB-SLAM2 [84]. A constraint based on epipolar geometry improves the labeling. SGC-VSLAM decrease the **Root Mean Square Error (RMSE)** of the ATE by 96% compared to ORB-SLAM2 [84] in highly dynamic environments. However, similar to DynaSLAM [85], its performance decreased in lower dynamics. Finally, Xing *et. al.* [103] proposes the DE-SLAM to deal with **Short-Term Dynamics (STD)** and **Long-Term Dynamics (LTD)** at the same time. A MobileNetv2 identifies bounding boxes of movable objects (cars, persons, *etc.*) classified as STD. A motion check of STD elements recognizes all moving objects in the current keyframe. As for LTD, DE-SLAM uses **Histogram of Oriented Gradients (HOG)** features extracted from ORB keypoints to improve its invariance to illumination changes. In the experiments, DE-SLAM [103] improved the localization over ORB-SLAM2 [84] in a changing environment. All of these methods using geometric constraints to improve semantic labeling only use static features for localization and mapping.

2.3.4 Prediction modeling

In the included works, Markov processes are used to predict the dynamics of the environment. Tipaldi *et. al.* [41] uses a dynamic occupancy grid and exploits the stationary distribution and the state holding time associated with **Hidden Markov Models (HMMs)** on a 2D grid. The method uses past observations for each run to learn the state transition probabilities iteratively to estimate the HMM parameters. Then, the localization can infer how often is expected to see a dynamic object in the environment and for how long. Comparing the proposed HMM-based localization to MCL using a standard grid, the former had a lower localization failure rate than MCL, capable of dealing with high dynamics (moving cars) and lower ones (parked cars). Rapp *et. al.* [42] implements a semi-Markov process extended by a Levy process to model a time dependency on the state holding time of Markov processes, also predicting as Tipaldi *et. al.* [41] the expected retention time for each cell being in a specific state. In the experiments, the proposed model [42] integrated in MCL improved the classic MCL in a dynamic environment.

The environment dynamics can have periodic patterns associated with them. Assuming periodic changing patterns, Krajník *et. al.* [43] proposes **FreMEEn** to model the probability of occupancy or feature visibility in a grid as a combination of harmonic functions related to periodic processes. FreMEEn [43] uses spectral analysis (Fourier transform) to compute the harmonic functions and predict future state with a given confidence. In a changing environment, FreMEEn [43] outperformed a static map in terms of localization error by selecting the most likely visible features at each location for localization. Santos *et. al.* [104] adopts FreMEEn [43] within an exploration scheme, where the planner predicts which areas are more likely to change at a certain time and generate the subsequent locations to explore. The experimental results showed that considering the environment dynamics increases the amount of information gathered compared to static models. Unlike FreMEEn [43], Wang *et. al.* [44] models both aperiodic and periodic changes by an **ARMA** model. This model describes

time series as stationary stochastic processes in terms of polynomials. While **FreME**n [43] is able to update recursively its model online, **ARMA** [44] only is updated once a day based on past observations. However, the model [44] achieved a higher prediction accuracy than **FreME**n [43] and a lower localization failure rate than both **FreME**n [43] and Tipaldi *et. al.* [41].

Instead of modeling the dynamics in the map, Thomas *et. al.* [81] uses a KPConv network to predict online dynamic motion labels of points with single 3D laser scans as input. The method [81] is a self-supervised learning approach with two main modules: PointMap and PointRay. PointMap is an **ICP**-based **SLAM** algorithm to provide a point cloud map for the annotation process. PointRay uses a similar approach to Pomerleau *et. al.* [93] to approximate ray tracing using spherical coordinates for obtaining the training annotation of dynamic labels: permanent (static points over all sessions), ground (to avoid ray tracing ground samples), and long-term (still objects in single sessions but relocated between sessions) and short-term (dynamic objects) movables, with the localization not considering the latter two. In the simulation experiments, PointMap with the proposed prediction module led to lower localization pose errors than the standard **MCL** algorithm.

2.4 Conclusions

The systematic review on long-term localization and mapping (available in Appendix A) discusses literature works on the topics of appearance invariance to changing conditions, dynamics modeling, map sparsification, and multi-session, among other computational issues, allowing the identification of current challenges in the literature and future directions for long-term **SLAM**. Given the focus of this research proposal on **SLAM** in dynamic environments, the current chapter focuses on literature works related to dynamics modeling found following the systematic methodology described in Appendix A and summarized in Section 2.1.

Next, the literature analysis of dynamics-related works presented in Section 2.3 discusses map representations proposed in the methodologies for modeling dynamics, dynamics identification and modeling using map matching, semantic detection of dynamic objects, and prediction of the environment dynamics. The environment representations proposed in the literature to handle moving elements in the scene are **STM/LTM** ([5, 30, 31]) memories and static/dynamic maps ([24, 34, 35]), and a multi-hypothesis framework ([79]). The first two techniques are very similar. Indeed, the **LTM** memory tries to filter dynamic data in the mapping process to only keep the most persistent and static observations in the map, similar to a static map. As for **STM**, this memory only keeps temporary and dynamic observations, equivalent to a dynamic representation of the environment. However, these map representations formulate assumptions on the environment dynamics, such as the number of consecutive observations to be considered a static observation ([30, 31]), certain semantic classes assumed to be dynamic observations ([34, 35]), or discrete timescales considered in the mapping process ([5]), or are susceptible to false positives and negatives ([24]), decreasing the accuracy of the environment representation. As for a multi-hypothesis formulation, while allowing recoverability from unmatched observations to the map, the limited computational resources on an **AMR** restrain the number of different hypotheses; plus, the proposed framework in [79] is very dependent on the uniqueness of the observations' signatures, which could not be valid for wide spaces and several rooms with similar appearance/structure. In terms of map matching, ray tracing-based techniques ([92–94]) exploit the free space information to identify dynamic changes between different observations. However, these techniques formulate a binary classification problem (static or dynamic) and are not suitable for low dynamics (also known as semi-static, *i.e.*, changes such as structural or non-temporary

ones), the latter due to assuming right way a change as a dynamic observation. Algorithms focused on the differences between the observations between the sensor data and the map ([25, 78, 82, 86–88]) are also focused only on the binary classification problem to reject dynamic changes for the localization process. Although the hybrid representation of **NDT** with occupancy grid maps ([95–97]) is required to model free space compared to the standard **NDT** formulation, the models used in those works favor newer observations, limiting the environment dynamics perceived by the algorithms.

Furthermore, the dynamic objects detection algorithms found in the literature focus on retrieving semantic data from the sensor observations, independently of being bounding box detection and classification ([35, 37, 102, 103]), or point-cloud ([98]) and image pixel-wise segmentation ([34, 38–40, 80, 85]). The main drawback of object detection-based evaluation of the environment dynamics is the prior assumption of certain classes being dynamic. This assumption may not stand valid in the long-term, *e.g.*, when an object presumably dynamic remains static (and could be used for localization temporarily) or, on the contrary, a presumably static element is moving in the scene. As for prediction of environment dynamics, those algorithms use mostly Markov-based models ([41, 42]), evaluate and predict the periodicity of changes ([43, 44, 104]), or use **CNN**-based prediction ([81]). Markov-based algorithms estimate the expected retention time for each cell of a 2D grid, inferring how often and for how long a dynamic observation is predicted to appear in the cells. However, this inference is based only on a model computed offline (based on previous measurements) due to the computational requirements of estimating the models, not integrating the most recent sensor observations. Although **ARMA** [44] improves over **FreMEn** [43, 104], the former only updates offline its prediction model. Finally, the **CNN**-based approach [81] assumes a movable dynamic motion label for the points of single 3D **LiDAR** scans that could still be considered in the localization while in its temporary static state. Also, the learning process is only executed offline, even though uses ray-tracing for data labeling in a self-supervised way.

Overall, there are still open challenges in the literature on long-term **SLAM** in dynamic environments. Even the definition of dynamic itself, in terms of observations of the perceived environment by a mobile robot, is not clear in the literature. Indeed, what defines a dynamic, a semi-static, or a static observation? So, based on the literature discussion presented in this chapter and also on the systematic review available in Appendix A, the following items are considered to be open challenges for handling dynamic elements in the scene:

- **heterogeneous sensor configurations:** current literature on long-term **SLAM** focuses mostly on a single type of sensors and/or data;
- **multi-modal maps:** currently, only a single type of map (*e.g.*, grid, point cloud, feature) is considered in the mapping process when handling moving elements in the scene;
- **localization and mapping tightly-coupled with dynamics estimation:** instead of executing those processes independently, perform a joint estimation for the three processes;
- **online estimation of the observations dinamicity:** online determination of a dynamics value for the sensor observations;
- **human-machine interfaces:** interaction with users of **AMRs**, *e.g.*, for providing prior information provided by the user on the environment dynamics for localization and mapping tasks.

Lastly, the problem of lifelong localization and mapping in dynamic environments requires a **SLAM** framework. From the ones discussed in Section 2.2, GraphSLAM seems to obtain the best mapping accuracy, compared to scan matching or particle filter-based formulations [68], and to be the one most used recently, as noted in the data extraction process of the systematic review on long-term SLAM (see Appendix A). However, further discussion will be presented in the following chapters on how and which **SLAM** frameworks

this research proposal will consider for its research plan. Also, all implementations mentioned in Section 2.2 (HectorSLAM [57], GMapping [67], and SLAM Toolbox [74]) are open-source and available as ROS [75, 76] packages (`hector_slam`¹, `gmapping`², and `slam_toolbox`³, respectively), facilitating possible testing and comparison during the execution of the research plan.

¹https://wiki.ros.org/hector_slam

²<https://wiki.ros.org/gmapping>

³https://wiki.ros.org/slam_toolbox

Chapter 3

Thesis Statement

This chapter describes the goals of this PhD research proposal according to the problem statement under study. Thus, the chapter is divided into four sections. First, Section 3.1 contextualizes the problem under study. Next, Section 3.2 describes the problem that is the focus of the research proposal, defining the problem statement. Section 3.3 presents the research questions and hypothesis that will be considered throughout the execution of this research proposal. Then, Section 3.4 defines the aim and goals of this PhD proposal.

3.1 Problem Context

Autonomous Mobile Robots (AMRs) have been used for automating tasks, such as transportation and security monitoring, in logistic and retail companies, healthcare institutions, warehouses, agriculture businesses, among others [1]. Although the application of **AMRs** for those use cases can be in the context of indoor and outdoor environments, this research proposal focuses on the localization of ground **AMRs** indoors.

Moreover, an **AMR** requires a representation of its surroundings to localize itself relative to the indoor environment. The robot's sensors allow the perception of the environment and extraction of meaningful data for the consequent building of the environment representation [3]. The sensors may be proprioceptive or exteroceptive. The first ones measure internal values of the system (robot). Examples of proprioceptive sensors are the wheel encoders to measure angular speed of the wheels and **Inertial Measurement Units (IMUs)** for estimating 6 **Degrees of Freedom (DoF)** of the vehicle – position (x, y, z) and orientation (roll, pitch, yaw) – fusing the measures obtained from accelerometers, gyroscopes, and possibly magnetometers. Exteroceptive sensors acquire information from the environment. Examples of these sensors are 2D/3D lasers and RGBD cameras for obtaining range and bearing readings and cameras for vision. Although exteroceptive sensors are the ones that extract data from the environment, proprioceptive readings allow continuing the estimation of the robot localization while not receiving valid data from exteroceptive sensors. Also, the data from both types of sensors may be fused to improve even further the robot localization and mapping processes [3].

Thus, this PhD research proposal considers a mobile platform with both proprioceptive and exteroceptive sensors. The proprioceptive sensorization considered throughout the work plan execution will be dead reckoning sensors, able to estimate the relative motion of the vehicle. The data fusion of more than one sensor increases the redundancy of the dead reckoning pose estimation for the robot. For example, while wheel encoders only provide a 3 **DoF** estimation, an **IMU** sensor is able for estimating the 6 **DoF** of the robot pose,

useful not only for inclined indoor floors but also for detecting wheel slippery. As for the exteroceptive sensors, this PhD thesis will focus on range-bearing data obtained from sensors such as **LiDAR**, RGBD (the depth readings), or **Time-Of-Flight (TOF)** sensors. This choice justifies by the fact of range and bearing readings being less affected by appearance variance of the environment (*e.g.*, lighting conditions, weather and seasonal changes), especially indoors, focusing this research work on dealing with changing elements in the scene.

3.2 Problem Statement

As discussed in Section 1.1, a map does not require future updates in static scenes due to the map remaining consistent with the explored environment. However, most application use cases of **AMRs** have moving elements in the surroundings or reconfiguration of the environment that affect the consistency of the map throughout time. Indeed, the system should decide how and when to update the map, the so-called stability-plasticity dilemma, where a long-term **SLAM** system should adapt to new changes and preserve older ones, the latter in case the environment returns to older states [5].

This PhD research proposal focus on the problem of localizing a mobile robot within a dynamic indoor environment. This problem is directly related to **Simultaneous Localization and Mapping (SLAM)**, given the need to update the map concurrently to the localization upon dynamic changes in the surroundings of the robot. Furthermore, this research proposal is not limited to a specific type of dynamics, for example, only detecting semi-static dynamics – changes between running sessions, *i.e.*, changes occurring outside of the robot sensors' **Field-Of-View (FOV)** at different time instants that, when observed by the sensors, appear static but elements were effectively moved in the scene – or only dealing with highly dynamic elements in the scene (*e.g.*, moving persons or objects). Still, the assumptions on the sensorization setup discussed in Section 3.1 are considered in the proposal to focus to scope of the problem on the dynamics challenge of performing long-term **SLAM**, not on appearance variance issues that may also occur, as explained in Section 1.1.

A mathematical formulation for the problem of performing lifelong **SLAM** in dynamic environments that this research proposal aims to solve is presented in Equations 3.1 and 3.2. These formulations are similar to the ones shown in Equations 2.2 and 2.4 for the generic full **SLAM** problem and the same problem with unknown correspondences, respectively, the latter in case of feature-based localization and mapping. Those two equations formulate the posterior probability over the robot path ($x_{1:t}$) along with the environment map (m) and the correspondence variables ($c_{1:t}$) between the perceived features and the map landmarks, provided the control variables ($u_{1:t}$) and the sensor measurements ($z_{1:t}$). One difference in the formulation presented in this chapter (Equations 3.1 and 3.2) to the original formulation of the full **SLAM** problem is considering another unknown variable, specifically, the dynamics of the environment map ($d_{1:t}$). The other difference is wanting to estimate the environment map at different time instants ($m_{1:t}$), considering the context of the environment changing throughout time. Also, the formulation of the problem statement for online **SLAM** is presented in Equations 3.3 and 3.4, integrating out past posterior probabilities from the full **SLAM** problem.

$$p(x_{1:t}, m_{1:t}, d_{1:t} \mid z_{1:t}, u_{1:t}) \quad (3.1)$$

$$p(x_{1:t}, m_{1:t}, c_{1:t}, d_{1:t} \mid z_{1:t}, u_{1:t}) \quad (3.2)$$

$$p(x_t, m_t, d_t \mid z_{1:t}, u_{1:t}) \quad (3.3)$$

$$p(x_t, m_t, c_t, d_t \mid z_{1:t}, u_{1:t}) \quad (3.4)$$

3.3 Research Questions and Hypotheses

Given the problem discussed in Section 3.2, some research questions can be formulated, aiming a solution for the problem. The three questions presented in the present section are the ones considered for the execution of this PhD research proposal. For each research question, at least one hypothesis is formulated to guide the development of a solution to solve the problem of localizing an **AMR** in a dynamic indoor environment.

Additionally, the block diagram representative of this PhD research proposal and illustrated in Figure 3.1 supports the hypotheses formulation for the respective research questions. This diagram represents the case in which a robot is moving and interacting within a non-static real world environment. The robot's internal readings provide proprioceptive readings from dead reckoning sensors, allowing an estimation of a robot "state" that represents the estimated relative motion, among other variables. Although the exteroceptive sensors are also present on the robot, these sensors usually retrieve information from the environment, not from the robot itself. The information may be processed for feature-based **SLAM** implementations to generate the observed features, be sent directly to the localization and mapping module as raw data in the case of, *e.g.*, grid-based mapping, or even an hybrid implementation requiring both types of information for obtaining feature and grid maps. Then, the **SLAM** module estimating the localization of the robot and the mapping of its surroundings concurrently processes the robot state and the raw data of the sensors and/or features' state (*e.g.*, position and descriptor/signature) extracted from that data. The localization estimator may require the following modules: *Data association* (in the case of feature-based **SLAM**, to compute the correspondence variable), *Matcher* (computation of a homogeneous transformation between observed data and estimated map), *Pose Estimation* (robot pose in the estimated map), and *Loop Closure Detection* (recognition of a previously mapped location). As for the mapping process, for grid-based implementations, only the *Map Update* would be required, *i.e.*, update the map with the current observation, given the estimated robot pose. This update may not only include the insertion in the map of the observation but also optimize constraints between poses, the latter in the case of pose graph **SLAM**. For feature-based **SLAM**, the following modules should be considered in addition to the *Map Update*: *Landmark Tracking* (update the pose of the map landmarks, *i.e.*, the observed features mapped by the **SLAM** module), *Landmark Detection* (detect if the observed features are not present in the map and may become a map landmark), and *Landmark Initialization* (insert the landmark in the map representation). The added modules to the standard **SLAM** system represented inside the RicoSLAM system are relative to this research proposal and will be explained in the hypotheses for the research questions.

Research Question 1. *How to estimate the extrinsic parameters of the robot's sensors without external equipment to improve the data consistency between the sensors and, subsequently, the **SLAM** results?*

Although estimating the extrinsic parameters of the robot's sensors is not related directly to the **SLAM** problem, fusing the data from multiples sensors for the localization and mapping modules require processing the sensors data in a common coordinate frame, usually, the robot frame. The data fusion process needs to know

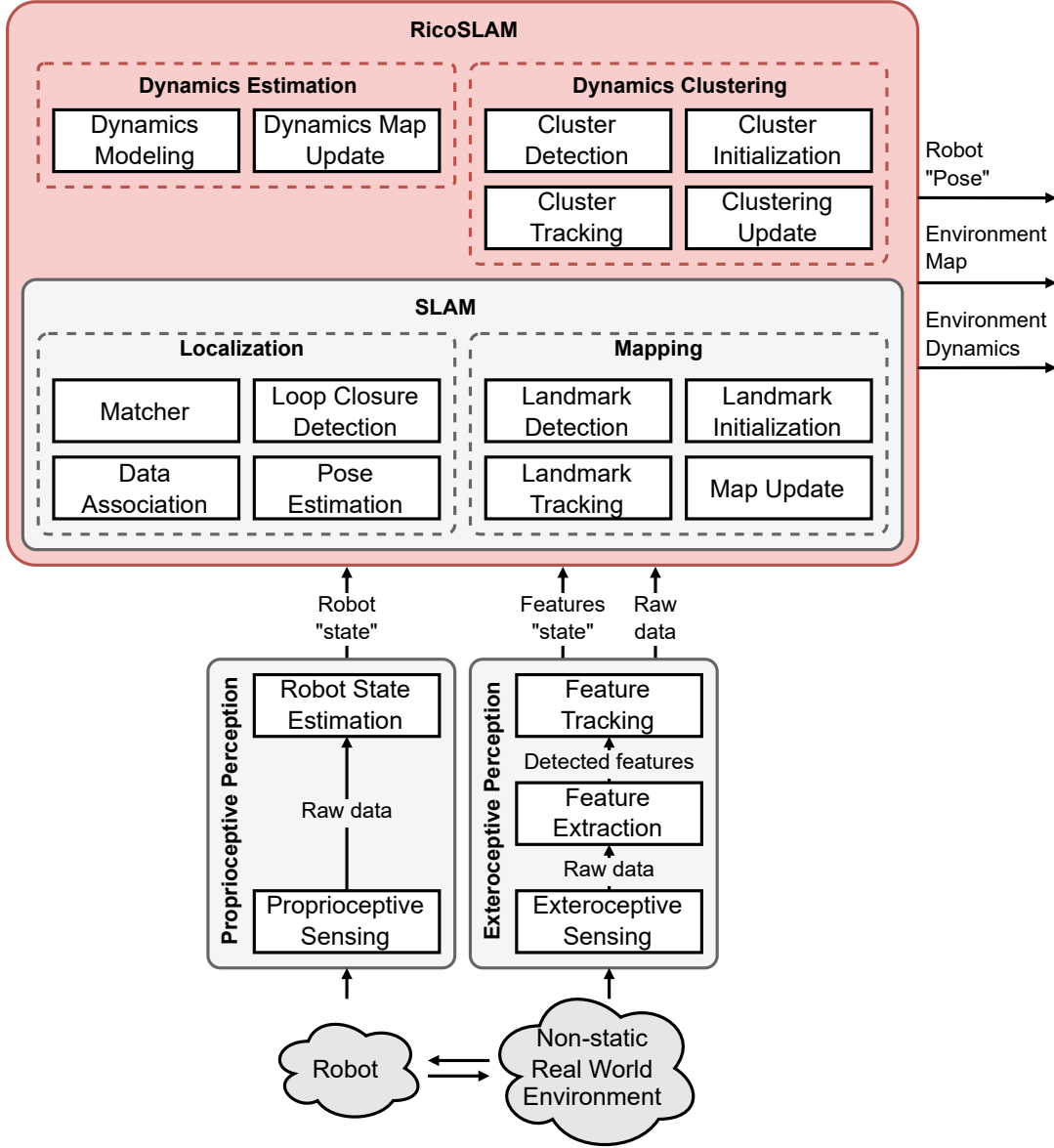


Figure 3.1: Generic block diagram for this PhD thesis research proposal in the context of **SLAM**.

the value of the extrinsic parameters of each sensor, *i.e.*, the 6 **DoF** of the homogeneous transformation that relates the sensor's coordinate frame to the robot one. Subsequently, errors relative to the extrinsic parameters may lead to inconsistencies between the data from different sensors, inducing errors into the **SLAM** system.

Previous studies from Sousa *et. al.* [105, 106] identify the need for calibration methods to compute the extrinsic parameters of the sensors relative to the robot frame, underlining the importance of the Research Question 1. For answering the latter question, this research proposal formulates the Hypothesis 1.1 focusing on an automatic approach that do not require external systems to the robot, only the proprioceptive and exteroceptive sensors present on the autonomous system. This approach takes on the basis that the egomotion (*i.e.*, relative motion estimated by a sensor) estimated from different data sources of a common coordinate frame (*e.g.*, the robot one) should converge when the extrinsic parameters are equal to their true value.

Hypothesis 1.1. *The maximization of the convergence of relative motion estimations of the robot's different sensors in the same coordinate frame, when actuating the robot to follow not necessarily a pre-defined path but*

one that can make observable all the extrinsic sensor parameters, will lead to an improve consistency on the data estimations of the sensors when performing SLAM with multiple sensors, including both proprioceptive and exteroceptive data sources.

Research Question 2. *How to improve the mapping results and their long-term stability when performing SLAM with a ground mobile robot in a dynamic indoor environment?*

In the literature (reviewed in Chapter 2), the existent methodologies for long-term SLAM either classify the dynamic environments into static, semi-static, and dynamic, or estimate the dynamicity of the observations offline. The consideration of the environment dynamics for the mapping process is usually performed by a binary classification, where only the static and long-term changes are added into the map representation in the robot used for localization. Furthermore, the concept of environment dynamics is not clearly defined in the literature in terms of which information retrieved from the environment estimates the dynamics.

This PhD research proposal formulates the Hypotheses 2.1 and 2.2 aiming a solution for the Research Question 2, hypothesizing on how the estimation of the environment dynamics can improve the mapping process in a dynamic environment. The first hypothesis states that the estimation of environment dynamics should account the map area (*e.g.*, the history of the dynamics in a certain area, the interaction of a user with the system to define different initial guesses of the dynamics in different areas), the velocity and position of map points, and whether the points exist or not throughout time when estimating the dynamics, while also considering the dynamics clustering properties of map points relative to their neighborhood (the points tend to have the same dynamic properties as the neighbor ones [88]). Then, the second hypothesis formulates that the estimated environment dynamics can improve the long-term stability of the mapping results and their consistency to the environment by giving more importance to static elements perceived in the scene. However, this importance does not ignore nor exclude all dynamic and changing elements, due to some of them may be useful for the localization process. The Hypotheses 2.2 and 3.1 (the latter will be discussed in the context of the Research Question 3) are directly related to each other. The reason is that those two hypothesize on the tightly-coupling of the localization and mapping process with the dynamics estimation of the environment to improve the long-term results of continuously performing SLAM.

Hypothesis 2.1. *The study of the dynamics history in the area to update the environment representation, the perceived and mapped points state (position, velocity), and if those points exist or not throughout time, while also hypothesizing on the dynamics clustering properties of the same points relative to their neighborhood, will give a model of the environment dynamics.*

Hypothesis 2.2. *The tightly-coupling of the environment dynamics modeling into the mapping process, using the estimation of the dynamics to weight the relevance of the map points for the environment representation, while also rejecting highly dynamic observations and giving more importance to static ones, will improve the long-term stability of the mapping results and their consistency to the environment most structural and static elements when performing SLAM with a ground mobile robot in a dynamic indoor environment.*

Research Question 3. *How to integrate the environment dynamics modeling in the localization process when performing SLAM with a ground mobile robot in a dynamic indoor environment?*

As already discussed in Chapter 2, most of the existent methods only use the static elements represented in the map for localization. Those methods also use the binary classification on the environment dynamics to

decide which elements to the map. However, there are elements in the scene that are can be dynamic but, for the time being, remain static and could be used for localizing the robot in the environment.

So, the formulation of the Hypothesis 3.1 aims to answer the Research Question 3 on how the estimated environment dynamics may help localizing the robot in a dynamic environment. This hypothesis specifies that the estimated dynamics for the map points can be weights for each observation in the matching process, while also considering the approach where some of the observations may be completely ignored. The goal is to give higher importance to completely static objects but not ignoring all dynamic elements present in the map, due to some of them may improve the localization estimation of the robot.

Hypothesis 3.1. *Similar to the mapping process, the tightly-coupling of the environment dynamics modeling into the localization of a ground mobile robot, considering the map points dynamics as weights in the data association and/or map matching processes, while also completely rejecting highly dynamic observations, will achieve long-term stability of the localization results in a dynamic indoor environment.*

In relation to the block diagram presented in Figure 3.1, this PhD research proposal intends to develop at least the following modules:

- *Dynamics Estimation*: compute the dynamics of the environment;
 - *Dynamics Modeling*: consider the dynamics history, the observed and mapped points state, if the points exist or not, and the clustering properties of neighbor points;
 - *Dynamics Map Update*: update the dynamics associated to the points of the map representation of the environment.
- *Dynamics Clustering*: cluster the map points accordingly to their estimated dynamics
 - *Cluster Tracking*: update the pose of the environment dynamics clusters;
 - *Cluster Detection*: detect if map points not associated to a cluster may then belong to one;
 - *Cluster Initialization*: initialize new clusters based on the dynamics properties of the map points;
 - *Cluster Update*: update the dynamics state of the clusters.

Both *Dynamics Estimation* and *Dynamics Clustering* modules are directly relative to Hypothesis 2.1, while the module *Dynamics Map Update* that outputs the estimated dynamics of the environment is also related to Hypotheses 2.2. As for the contribution relative to the Hypothesis 1.1, improving the accuracy of the extrinsic sensor parameters will affect the *Robot State Estimation* and *Feature Extraction* modules.

3.4 Aim and Scope

Overall, the main objective of this PhD proposal, *RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments*, is to research on a localization and mapping system able to localize successfully a mobile robot in a dynamic environment, performing lifelong **SLAM**. In order to reach the primary goal, this research thesis proposes to study, benchmark, and/or research on the following items:

- automatic calibration methods for estimating the extrinsic parameters of sensors on a mobile robot;
- sensor models compatible with a **SLAM** probabilistic framework to account the environment dynamics;
- modeling environment dynamics of map points as a continuous value;
- clustering algorithms for evaluating the dynamics of map points contemplating the value from neighbor points;

- data association algorithms for feature-to-landmark and point-to-cluster associations accounting the environment dynamics;
- matching algorithms using raw data, extracted features from the observations, and/or hybrid approaches of both feature and raw data implementations for considering the environment dynamics;
- map update methodologies to model the environment dynamics into the map and considering them when adding and removing single points and/or clusters of points from the map representation.

Nevertheless, this research work will focus on developing algorithms capable of localizing mobile robots in a dynamic indoor environment under real industrial scenario applications.

Chapter 4

Preliminary Developments

Next, this chapter presents preliminary works preceding the ones planned and discussed in this PhD thesis research plan (see Chapter 5). First, Section 4.1 explains the modifications performed on mobile robotic platforms to accommodate different sensors for use throughout the work plan execution. Section 4.2 discusses preliminary research results relative to testing Hypothesis 1.1 presented in Section 3.3. Then, Section 2.2.1 presents the initiation to the SLAM problem in a real scenario.

4.1 Preparation of the Mobile Platforms

The goal of this PhD research proposal is achieving long-term localization and mapping in dynamic environments. Although the use of simulation environments can be useful to start the development of the required algorithms to achieve that goal, the simulation may not represent all the challenges that a real changing environment poses to performing SLAM, such as sensor noise, robot vibration when moving passed to the sensors, arbitrary and/or unexpected changes in the scene, sensor obstructions, among other factors.

Consequently, some preliminary works preceded the elaboration of the research plan to set up experiences in real scenarios and prepare the robots. Figure 4.1 presents two mobile platforms present in room I-108 at the Faculty of Engineering of the University of Porto (FEUP). The first platform (Figure 4.1a) is the five three-wheeled omnidirectional robots belonging to the 5dpo robotics team from FEUP¹. The modifications made to this robot are the following ones: stabilization of the firmware (communication between the hardware and software), development of ROS [75] drivers to allow the motion control of the robot (the software present in the robot is a Lazarus²-based distributed architecture), interface of the robot for interacting with the INESC TEC – Institute for Systems and Computer Engineering, Technology and Science Robotics Navigation stack (*e.g.*, allow the navigation in an existent graph), and design and print of a 3D part for allowing setting up different sensors on the top of the robot’s “head” (originally, the robot used a catadioptric monocular vision system to obtain a 360° FOV). The latter modification shown in Figure 4.1a demonstrates the mounting of a Hokuyo UST-10LX 2D laser sensor. However, the model can be adapted to allow other sensors on the robot, such as a RoboSense RS-Helios 5515 3D LiDAR or a OAK-D RGBD sensor (available at INESC TEC). As for the other platform (see Figure 4.1b), there is available two Hangfa Discovery Q2 four-wheeled omnidirectional robots

¹<https://web.fe.up.pt/~robosoc/>

²<https://www.lazarus-ide.org/>

owned by **INESC TEC** in the **FEUP**'s I-108 room. The modifications on this robot are mostly related to changing the original proprietary motion drive system to a similar hardware architecture of the 5dpo's three-wheeled robot. Those hardware modifications are required to obtain more control of the platform on the firmware and software sides. Further changes to the Hanfga platform are foreseen relative to a mounting system to allow different types of sensors for data acquisition, similar to 5dpo's robots. Also, the fact that both platforms shown in Figure 4.1 have an omnidirectional steering geometry makes them holonomic robots, *i.e.*, they do not have non-holonomic kinematic constraints, being able to move in any direction in space at any time [3].

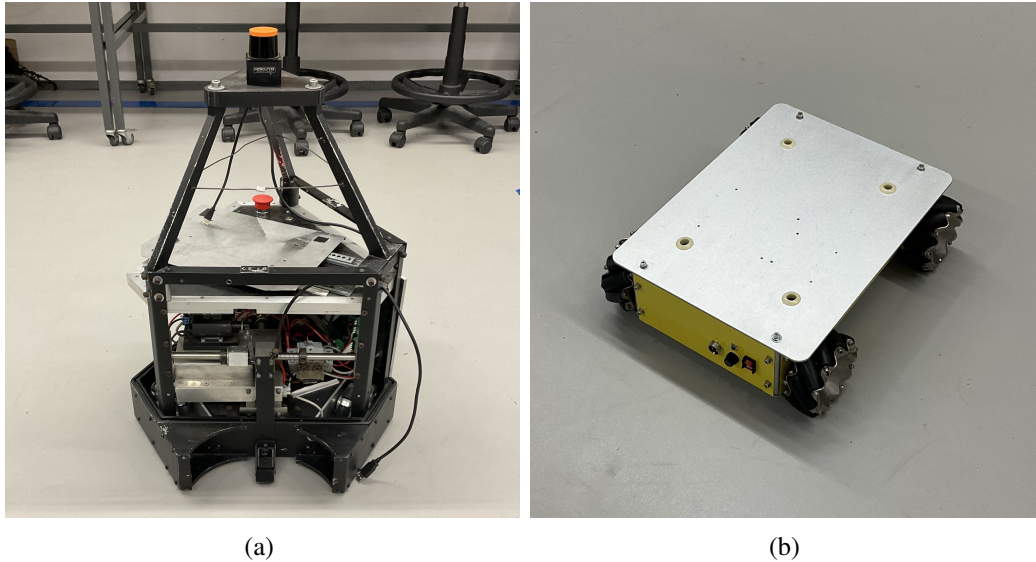


Figure 4.1: Ground mobile platforms available at the **Faculty of Engineering of the University of Porto (FEUP)** for this PhD thesis: (a) three-wheeled omnidirectional robot of the 5dpo robotics team; (b) Hangfa Discovery Q2 four-wheeled omnidirectional robot.

The installation of the two platforms shown in Figure 4.1 at **FEUP**'s facilities, in terms of them navigating autonomously, poses an interesting challenge in terms of environment dynamics. One example is the robots moving in the -1 floor of the **Department of Electrical and Computers Engineering (DEEC)** at **FEUP** between the room I-108 and the rooms I-113 and I-114 (see Figure 4.2), where the latter two are occupied by the **CRIIS – Centre for Robotics in Industry and Intelligent Systems** from **INESC TEC**. The halls have highly dynamic elements represented by persons moving in the scenario and doors. As for the rooms themselves, the I-108 has semi-static changes occurring between visits due to that room being shared by several research teams and also personnel from **FEUP**, involving boxes, chairs and other elements moving from place to place.

Furthermore, another mobile platform modified in the preliminary phase to the elaboration of this PhD research proposal is illustrated in Figure 4.3. The mobile platform is a tricycle robot installed by **INESC TEC** at the IDEPA's facilities³. Initially, the robot had a Hokuyo UST-10LX 2D laser at the top of it dedicated to autonomous navigation purposes, given that the higher the sensor is the scene is expected to be more static, considering a 2D perspective. However, the goal of this PhD thesis is to deal with changing environments. Thus, two Sick S300 2D lasers were installed near the ground to increase the environment dynamics captured by these two sensors (*e.g.*, legs from moving people, boxes changing places). A 360° FOV configuration (putting the sensors at opposite corners of the base) was not possible to install due to safety directives defined

³<https://www.idepa.com/>

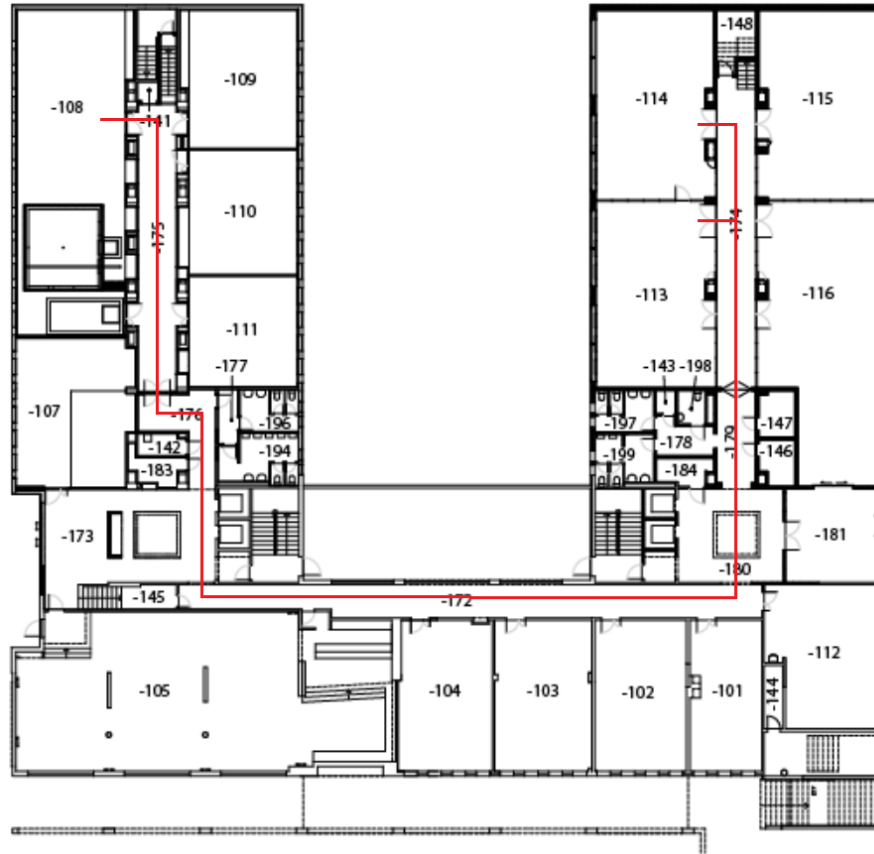


Figure 4.2: Floor -1 of the Department of Electrical and Computers Engineering (DEEC) at FEUP. Path between the room I-108 and the rooms I-113 and I-114 illustrated as a red line in the floor plan.

by the company (the sensors would stick out of the robot's base frame, possibly endangering the workers). Nevertheless, the 3D part designed and printed specifically for the IDEPA's robot allowed the installation of the two lasers at the front and the back of the vehicle, only not being visible to the sensors the environment at the sides of the robot. Also, the 3D support of the lasers allows adjusting them in terms of pitch and roll, facilitating the installation of these for trying to insure parallel readings to the ground.

The IDEPA's case study will be interesting for this PhD research proposal considering that it is a real scenario in an industrial environment. As discussed in the systematic review available in Appendix A, most of the public datasets related to the SLAM problem are mostly in laboratory, office, or outdoor environments. Therefore, the data acquisition occurring at the IDEPA's facilities will be interesting for testing and validating the results foreseen for this PhD thesis in an industrial context.

4.2 Robot Calibration

Next, in the scope of the curricular unit Advanced Topics in Electric and Computers Engineering, initial tests relative to research on Hypothesis 1.1 are presented in a technical report [107]. These tests followed an optimization-based formulation similar to GraphSLAM (see Section 2.2.4) for estimating the extrinsic sensor parameters of wheel odometry–2D laser sensor setup. The proposed methodology requires at least two virtual measurements (see Section 2.2.4) for each edge for evaluating the convergence between two different

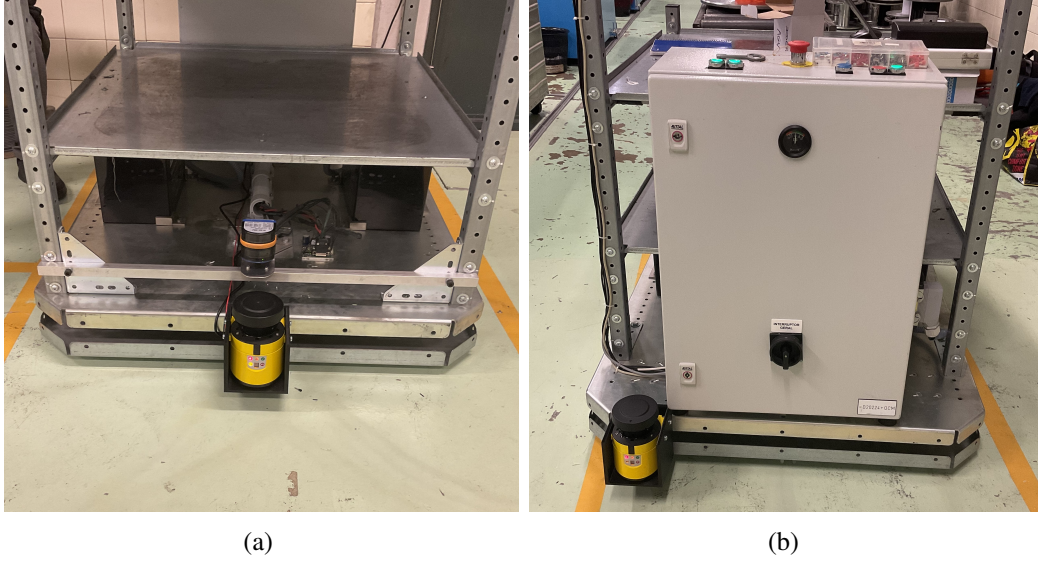


Figure 4.3: Tricycle ground mobile robot installed by the **INESC TEC – Institute for Systems and Computer Engineering, Technology and Science** at the IDEPA’s facilities: (a) front-view; (b) back-view.

sources (z_{ij}^{sens1} and z_{ij}^{sens2} , respectively), following the statement in Hypothesis 1.1. In the case scenario tested in report [107], one measurement z_{ij}^{odo} corresponds to the wheeled odometry estimation between pose x_i and x_j and the other measurement z_{ij}^{las} to the matching result between the 2D laser scans acquired at those same poses. Figure 4.4 illustrates how the edges are created in the report, considering the initial pose x_0 the starting end of all edges, and the sampled poses through the remaining of the robot path ($x_{1:N}$) at the other end of the edges.

Furthermore, the methodology followed in the report [107] considers the same calibration path as in the odometry calibration work of Sousa *et. al.* [108], given the circular path with decreasing and/or increasing radius demonstrated to make observable all odometry calibration parameters. However, the methodology is not limited to only that calibration path. As for the mathematical formulation of the optimization-based approach, Equation 4.1 defines the cost function to be minimized by an optimization algorithm. This equation differs from the one for GraphSLAM (see Equations 2.7 and 2.8) in the sense that is not the error between the virtual and the predicted measurement that will be minimized ($z_{ij} - \hat{z}_{ij}$), but instead, the difference between different virtual measurements along the same edge ($z_{0j}^{odo} - z_{0j}^{las}$). The formulation shown in Equation 4.1 considers that the virtual measurements relative to wheel odometry (z_{0j}^{odo}) and scan matching (z_{0j}^{las}) depend on the respective odometry (s^{odo}) and extrinsic 2D laser (s^{las}) parameters, the latter ones defining a 2D homogeneous transformation (2D translation and a yaw angle) relative to the robot’s coordinate frame.

$$\{s^{odo,*}, s^{las,*}\} = \underset{s^{odo}, s^{las}}{\operatorname{argmin}} \sum_{j=1}^N \|z_{0j}^{odo} - z_{0j}^{las}\|_2 \quad (4.1)$$

However, as the technical report indicates, several improvements can be made to the proposed automatic calibration methodology, some of them are the following ones:

- improve the matching result by extracting geometric features (lines, planes, corner points) from laser data instead of using only the raw data of the sensor;
- research on optimization algorithms able to improve the computational performance when searching a solution for the calibration parameters;

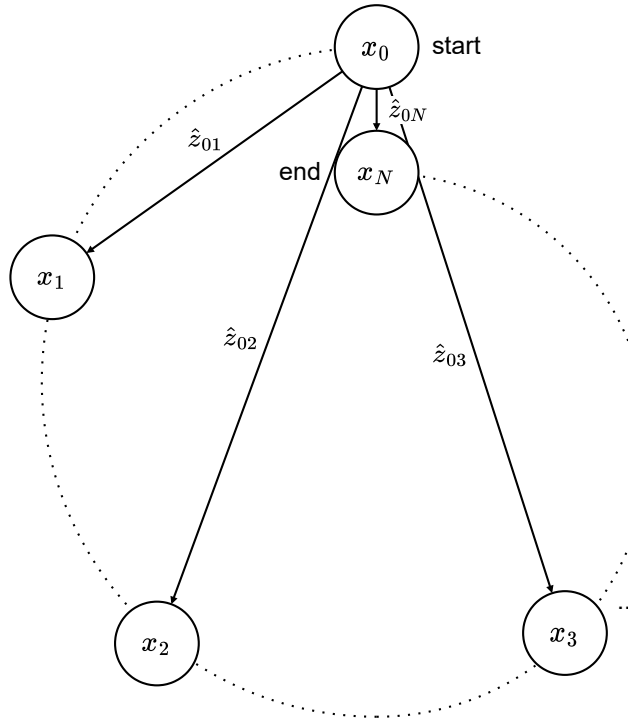


Figure 4.4: Creation of the edges represented by the predicted virtual measurements $\hat{z}_{ij} = x_j - x_i$ in the graph between robot poses x for the automatic calibration algorithm. The calibration path taken by the robot starts at pose x_0 and ends at x_N .

- testing the methodology with other sensors.

The improvements to be made to the methodology mentioned previously will be performed during this PhD thesis proposal, as will be discussed in Chapter 5.

4.3 Introduction to SLAM

Lastly, the other preliminary work preceding the elaboration of this research proposal is an initial mapping result obtained from using the ROS [75, 76] package `slam_toolbox` mentioned in Chapter 2. The environment considered for this first experience is room I-108 at FEUP, using the 5dpo team's three-wheeled omnidirectional robot for the mapping process. Also, the resulting grid map is illustrated in Figure 4.5, while the mapping process is available for illustration purposes in a YouTube video⁴.

The map shown in Figure 4.5 first validates the preparation work of the robotic platform from the 5dpo robotics team for use in this PhD thesis proposal in terms of the sensor combination wheeled odometry–2D laser. Moreover, the artifacts on the map near the room walls are relative to boxes, chairs, and mobile tables, among other changing elements. The presence of these elements shows why room I-108 is an interesting case study for the research on SLAM in dynamic scenes. Nevertheless, this PhD research proposal will carry out more experiences and research on the topic, as will be explained in the next chapter.

⁴<https://youtu.be/ax8cOVe6Z5I>



Figure 4.5: Map resulting from a **SLAM** experience using the `slam_toolbox` ROS package with the sensor setup wheeled odometry–2D laser on the 5dpo’s three-wheeled omnidirectional robot.

Chapter 5

Research Plan

The last chapter of this proposal details the PhD thesis research plan for achieving the goals established in Section 1.3 and solving the problem stated in Chapter 3. First, Section 5.1 describes the tasks foreseen for the execution of the research plan and presents the timeline. Section 5.2 defines milestones for the PhD thesis and the publications expected to be published within the scope of the thesis. Then, Section 5.3 explains the methodology for scientific dissemination of the main results of this research proposal.

5.1 Tasks and Timeline

This PhD thesis research plan will be executed in three years, starting on January 2023 and expected to end on December 2025. The host institution is INESC TEC. Furthermore, the proposed work is in line with the roadmap of its mobile robotics team for industrial applications, aiming for a solution of lifelong autonomy in dynamic environments. This work also benefits from a close relationship with the GreenAuto project from the European Union (EU) Recovery and Resilience Plan, where the project is related to research on autonomous mobile systems for logistics scenarios with flexible storage.

Furthermore, the complete timeline with all the tasks foreseen for the execution of this research proposal's plan is illustrated in Figure 5.1. The plan is composed of five main tasks: *T1 SLAM Framework* (research on SLAM solutions and on the modules that integrate the localization and mapping processes), *T2 Robot Calibration* (study the automatic robot calibration problem in the context of Hypothesis 1.1), *T3 Dynamics Modeling* (research on dynamics modeling methodologies to explore the validity of Hypotheses 2.1, 2.2, and 3.1), *T4 Test and Validation* (tasks related to the integration of the PhD thesis contributions), and *T5 PhD Thesis* (report the results of the thesis). Next, further detail is given for each subtask present in the plan.

T1 SLAM Framework

T1.1 Literature and commercial SLAM frameworks Study, research, and compare existing SLAM solutions in the literature and, if possible, commercial systems. This task will also identify the advantages, disadvantages, and possible limitations of existing solutions, while also defining adequate evaluation metrics for the comparison of SLAM frameworks.

T1.2 Feature extraction algorithms Study, research, and compare feature extraction algorithms. This task will also study feature extraction depending on different data sources (e.g., 2D and 3D LiDAR).

T1.3 Data association algorithms Study, research, and compare data association algorithms for estimating the correspondence variables between the observed features and the landmarks represented in the map.

T1.4 Sensor data matching algorithms Study, research, and compare data matching algorithms between the observed data and the current map representation of the environment. The observed data can be defined as the raw sensor data, extracted features, or even an hybrid approach.

T1.5 Feature versus grid-based localization Study, research, and compare feature and grid-based localization algorithms. This task will also consider the study and research on an hybrid approach, where both feature and grid data are considered simultaneous for estimating the robot's pose.

T1.6 Base SLAM framework for RicoSLAM Research on a base SLAM framework for this PhD research proposal. This framework and the respective architecture should consolidate all the knowledge obtained from the results of T1.1–1.5.

T2 Robot Calibration

T2.1 Geometric feature extraction and matching Study, research, and compare feature extraction and matching algorithms based on geometric cues from the sensors data (*e.g.*, corner points, lines, and planes). This task will possibly leverage the results from tasks T1.2 and T1.3, given the relationship of those two tasks relative to T2.1 in terms of using features for localization.

T2.2 Automatic calibration algorithm Research on an automatic calibration algorithm to test the validity of Hypothesis 1.1.

T3 Dynamics Modeling

T3.1 Environment dynamics as a continuous value Research on a model for estimating the environment dynamics as a continuous value and testing the validity of Hypothesis 2.1.

T3.2 Dynamics clustering algorithm Research on a dynamics clustering approach to cluster the map points accordingly to their environment dynamics properties and similarity to the points neighborhood. The approach should consider algorithms from task T1.3 and modify them for the update of the dynamics clustering map.

T3.3 Data association considering dynamics Research on a data association algorithm to consider the environment dynamics and improve the data association results.

T3.4 Map update accounting dynamics Research on a map update methodology for considering the estimated environment dynamics and testing the validity of Hypothesis 2.2.

T3.5 Localization accounting dynamics Research on a localization algorithm for considering the environment dynamics in the pose estimation and data association processes and testing the validity of Hypothesis 3.1.

T4 Test and Validation

T4.1 Integration and validation Consolidate, integrate, test, and validate the PhD contributions in the a **SLAM** framework able to deal with dynamic elements in the scene and achieve lifelong autonomy.

T4.2 Data acquisition Acquire data using different sensors (*e.g.*, 2D and 3D lasers, RGBD sensors) in adequate environment conditions (*i.e.*, ability of perceiving a variety of different dynamic conditions of the scene – highly dynamic elements, semi-static changes, *etc.*) in the scope of this PhD research proposal. This task includes the data acquisition experiences to be taken at **FEUP** (room I-108) and IDEPA (industrial application) facilities with the different mobile platforms considered for this PhD thesis (see Section 4.1).

T5 PhD Thesis

T5.1 Writing the thesis Report the thesis results obtained from all the other tasks of the research plan in the official document to be delivered at the end of the PhD.

T5.2 Scientific dissemination Write the main results of the PhD thesis in scientific publications. Disseminate the results in the scientific community using the appropriate channels for that effect (further details given in Section 5.3).

5.2 Milestones and Publications

Based on the timeline presented in Figure 5.1, this research proposal defines the following milestones:

- **M1 Automatic sensor calibration algorithm:** finish the research on the automatic sensor calibration algorithm to be used for estimating the extrinsic sensor and odometry parameters of the robot platforms to be used during the plan execution;
- **M2 Base SLAM framework for RicoSLAM:** deadline for the implementation of a base **SLAM** framework for this PhD thesis based on the results from task T1;
- **M3.1 Dynamics Modeling Estimator:** target date for finishing the research on a environment dynamics estimator to be used for the tasks T3.3–3.5;
- **M3.2 RicoSLAM:** deadline for the integration and validation of all contributions related to this PhD research proposal in the desired framework for performing lifelong **SLAM** in dynamic environments, the *RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments* framework.

This research proposal defines also the expected publications to be submitted for publishing the main results of this PhD thesis. The foreseen publications are the following ones:

- **P1 Benchmarking of existent SLAM solutions:** article reporting the comparison results not only from task T1.1 but possibly also from tasks T1.2–1.5;
- **P2 Automatic sensor calibration algorithm:** conference paper presenting the proposed approach for automatically estimating the extrinsic sensor parameters without recurring to an external system, based on the results from task T2;
- **P3.1 Environment dynamics modeling:** conference paper presenting the methodology for estimating the environment dynamics as a continuous value based on task T3.1;

- **P3.2 Map update accounting the environment dynamics:** article reporting the map update methodology considering the dynamics environment estimation and comparison to the standard update and other approaches based on task T3.4;
- **P3.4 RicoSLAM:** article reporting the main results of this PhD thesis and comparison with current solutions;
- **P4 Long-term SLAM dataset in a changing environment:** article presenting a dataset in a dynamic environment in an indoor scene based on the results from task T4.2.

5.3 Scientific Dissemination

Finally, the results dissemination of this PhD research proposal is expected to be through publications for validating the work with peer revision from the scientific community during the PhD. A first analysis on publication venues related to the topic long-term localization and mapping is already present in the systematic literature review on that topic, available in Appendix A. Considering that first analysis and that the scientific dissemination will consider both international conferences and prestigious international journals, this research proposal identifies a list for the dissemination of the achieved results. The list is the following one and considers venues indexed by several scientific databases, including Ei Compendex, Inspec, Scopus, and/or Web of Science:

- Conferences:
 - ACM/IEEE International Conference on Human-Robot Interaction (HRI)
 - European Conference on Mobile Robots (ECMR)
 - Flexible Automation and Intelligent Manufacturing International Conference (FAIM)
 - Iberian Robotics Conference (ROBOT)
 - IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)
 - IEEE International Conference on Robotics and Automation (ICRA)
 - IEEE International Intelligent Transportation Systems Conference (ITSC)
 - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
 - International Conference on Computer and Automation Engineering (ICCAE)
 - International Conference on Mechatronics, Robotics and Automation (ICMRA)
- Journals:
 - Advanced Robotics (Taylor & Francis)
 - Autonomous Robots (Springer)
 - Frontiers in Robotics and AI (Frontiers)
 - IEEE Robotics and Automation Letters (IEEE)
 - IEEE Transactions on Intelligent Transportation Systems (IEEE)
 - IEEE Transactions on Robotics (IEEE)
 - International Journal of Advanced Robotic Systems (Sage Journals)
 - International Journal of Robotics Research (Sage Journals)
 - Journal of Field Robotics (Wiley)
 - Journal of Intelligent & Robotic Systems (Springer)
 - Robotics and Autonomous Systems (Elsevier)

In addition to publish the results in scientific conferences and journals, this PhD thesis proposal also considers social networks to help spreading out the achieved results. More specifically, all scientific contributions

and achieved results related to the thesis will be posted in ResearchGate and LinkedIn. The first will be used mostly for sharing preprint versions of the submitted papers. As for LinkedIn, this social network will be used as a way to communicate the different achievements accomplished throughout the research plan execution, including publications and relevant results in terms of performing **SLAM** in dynamic environments and automatic calibration of the sensors on a mobile platform. The sharing of results will also consider the elaboration of demonstration videos.

This PhD research proposal also foresees the publication of datasets, as a result from task T4.2 (see Section 5.1). These datasets will be public and published in the **INESC TEC** research data repository¹. The main advantages of using the latter are the following ones: link the datasets to the host institution of the PhD thesis (*i.e.*, **INESC TEC**), the compliance of the platform to Open Data in a research context², and the expected longevity of accessibility for the public links given the use of CKAN³ (open source data management system) and the repository being hosted in **INESC TEC** servers (*i.e.*, not dependent on third-party services). Also, the dissemination of the corresponding results will be in the form of scientific publications, while also spreading out those publications in the social platforms ResearchGate and LinkedIn mentioned previously.

5.4 Summary

Overall, the work plan presented in this chapter covers the research on all hypotheses discussed and formulated in Section 3.3. This plan defines all the tasks, milestones, and publications foreseen for the execution of this research proposal, and also defines the respective timeline. However, the plan will be refined during the PhD to deal with unforeseen events and possible delays and/or scientific obstacles found through the plan execution.

¹<https://rdm.inesctec.pt/>

²<https://opendefinition.org/od/2.1/en/>

³<https://ckan.org/>

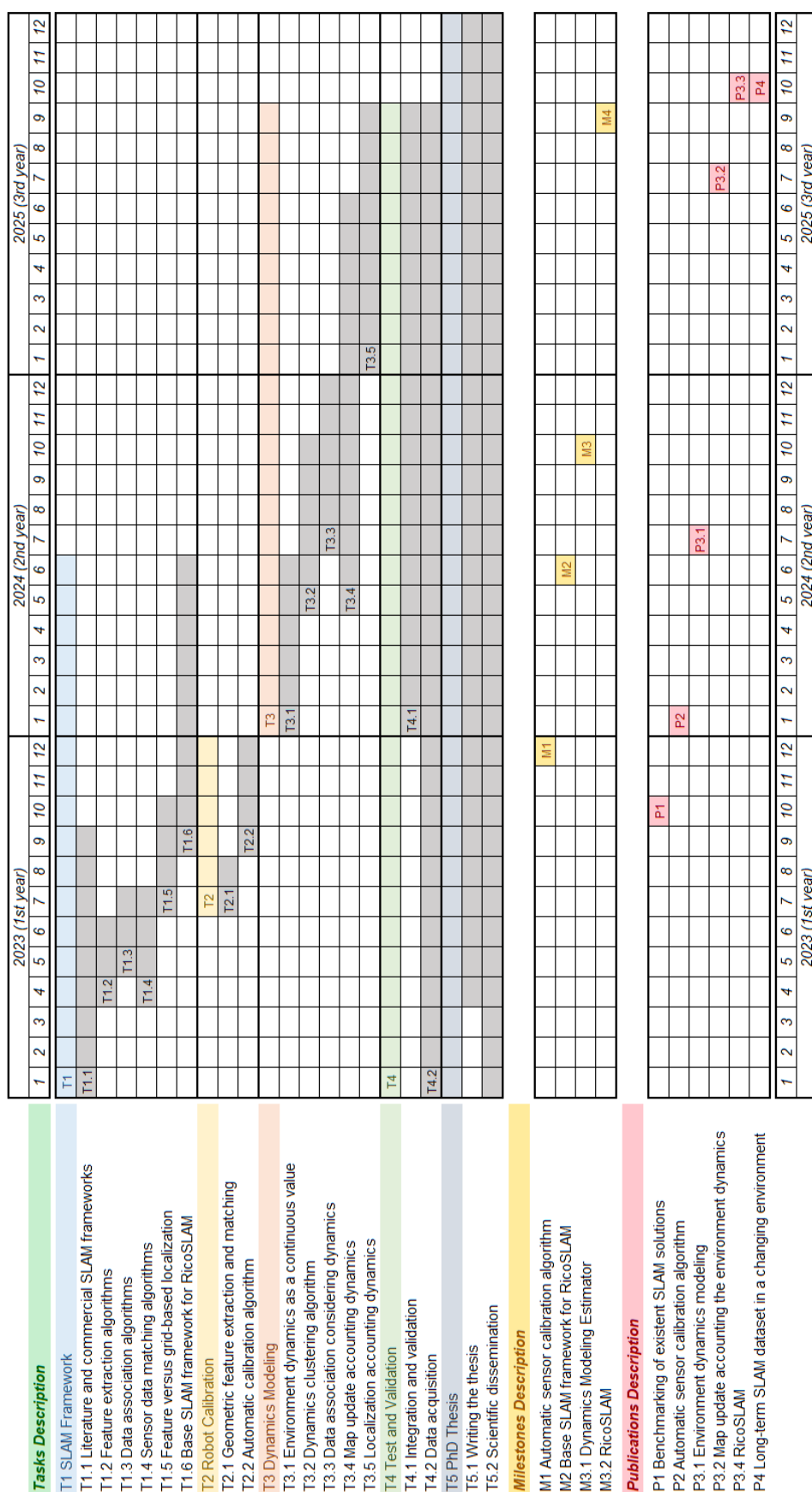


Figure 5.1: Timeline of the PhD research plan.

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Appendix A

A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots (preprint)

Ricardo B. Sousa, Héber M. Sobreira, and António Paulo Moreira. “A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots” [Manuscript submitted for publication]. 2022. Department of Electrical and Computer Engineering, Faculty of Engineering of the University of Porto (FEUP). Centre for Robotics in Industry and Intelligent Systems (CRIIS), INESC TEC – Institute for Systems and Computer Engineering, Technology and Science. URL: <https://github.com/sousarbarb/slr-ltln-mr/blob/main/preprint/pdf.pdf>